



**AI ASSISTED GRADING FRAMEWORK FOR THAI-LANGUAGE
WRITTEN EXAM QUESTIONS BASED ON LLM AND RULE-BASED
REASONING APPROACH**

BY

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ABSTRACT

Assessment plays a central role in education, providing insights into student comprehension, learning progress, and curriculum effectiveness. Written exams are particularly valuable as they capture higher-order cognitive skills such as reasoning, problem-solving, and expressive ability. However, manual grading of written responses is both time-consuming and prone to inconsistency due to factors such as evaluator fatigue, subjective interpretation, and variations in teaching assistant training. In the Thai context, these challenges are amplified by the language's complex grammar, tonal system, and context-dependent exceptions, which complicate evaluation and increase the likelihood of bias. Consequently, many educators rely heavily on multiple-choice examinations, which are easier to grade but insufficient for assessing deeper levels of understanding.

Recent advancements in Natural Language Processing (NLP), especially the development of Large Language Models (LLMs), provide new opportunities to address these challenges. LLMs offer sophisticated contextual analysis and semantic understanding that go beyond the limitations of traditional rule-based or keyword-based grading systems. However, their performance may still be unpredictable without structured guidance. This thesis proposes a hybrid AI-assisted grading framework that integrates the contextual reasoning

capabilities of LLMs with the transparency and consistency of rule-based reasoning. The framework is designed specifically for Thai-language written exam questions, aiming to reduce grading workload, enhance fairness, and support wider adoption of open-ended assessments in Thai education.

Through experimental evaluation, the framework demonstrates its ability to align closely with human grading while minimizing inconsistency. The results highlight its potential as a practical solution for teachers facing heavy workloads, as well as a step toward more equitable and comprehensive assessment methods in Thai education. This research contributes both a novel methodology for exam evaluation and empirical insights into the broader use of AI in educational contexts.

Keywords: Exam Evaluation, Natural Language Processing (NLP), Large Language Models (LLMs), Thai Language, Rule-Based Reasoning, Educational Assessment.

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LIST OF SYMBOLS/ABBREVIATIONS

Symbols	Terms
i	Index of the model (e.g., ChatGPT)
S_i	Normalized score for model performance (0–1)
R_i	Normalized reasoning quality score (0–1)
w_{Score}	Weight for performance score
$w_{\text{Reasoning}}$	Weight for reasoning score
WPS_i	Weighted Performance Score of model i
N_{match}	Number of exact score matches between raters or AI and human
N_{total}	Total number of evaluated items (questions)
SAR	Score Agreement Rate (%)
P_o	Observed agreement (proportion of agreement actually observed)
P_e	Expected agreement by chance
κ	Cohen's Kappa coefficient (inter-rater reliability)

Abbreviations	Terms
API	Application Programming Interface
AI	Artificial Intelligence
CASL	Contextual Access System for Login (JavaScript Library)
CAT	Category
CSV	Comma-Separated Values
ERD	Entity-Relationship Diagram
LLM	Large Language Model
LLMs	Large Language Models (plural)
MCP	Model Context Protocol
MCQ	Multiple-Choice Question
MTF	Multiple-True-False
NLP	Natural Language Processing
OEQ	Open-Ended Question
OpenFGA	Open Fine-Grained Authorization
PDF	Portable Document Format
TA	Teaching Assistant
TOPLA	LLM-TOPLA (Top-Layer Aggregation)
UI	User Interface
Teacher 1	Ms. Nabduan Phayom
Teacher 2	Ms. Tippanan Jankaew

CHAPTER 1

INTRODUCTION

Education relies heavily on assessment as a means of evaluating student learning, guiding curriculum development, and informing teaching practices. Among different assessment methods, written examinations stand out as a powerful tool for evaluating higher-order skills such as reasoning, problem-solving, and expressive ability. However, grading written exams poses significant challenges, particularly in contexts where linguistic complexity, large class sizes, and teacher workload combine to create barriers to fair and consistent evaluation. In Thailand, these challenges are even more pronounced due to the unique features of the Thai language and the increasing demand for efficiency in educational systems. This research addresses these issues by proposing an AI-assisted grading framework that leverages Large Language Models (LLMs) and rule-based reasoning to improve the evaluation of Thai written exam responses.

1.1 Background and Motivation

Examinations remain one of the most critical tools in educational assessment, serving as a benchmark for measuring students' comprehension, knowledge retention, and ability to apply concepts. Among different types of examinations, written exams are particularly valuable because they provide a deeper insight into students' higher-order cognitive abilities, such as critical reasoning, logical explanation, problem-solving, and language proficiency (Lee & Song, 2024). Unlike multiple-choice questions that primarily measure recognition and recall, written responses reveal students' thought processes, creativity, and capacity for structured argumentation.

Despite these advantages, written exam evaluation presents significant challenges. Traditional manual grading is time-consuming, resource-intensive, and often inconsistent. Human evaluators, including teachers and teaching assistants (TAs), are susceptible to fatigue, emotional states, and subjective interpretation, which may lead to inconsistency in grading outcomes (Månsson & Norgren, 2016). Even when grading rubrics are provided, variations in understanding and application among TAs may result in uneven assessment standards.

1.2 Challenges in the Thai Language Context

In the Thai educational context, the grading of written responses is further complicated by the unique linguistic and structural properties of the Thai language. Thai grammar is highly flexible, allowing for multiple valid sentence constructions. The language is tonal, meaning word meaning can change depending on pitch, and it contains numerous context-dependent exceptions (Tongsilp et al., 2024). Moreover, Thai writing omits spaces between words, which complicates segmentation and tokenization processes required for computational analysis.

These linguistic features make it particularly difficult for evaluators to interpret student responses accurately, especially when assessing reasoning, argumentation, or nuanced explanations. Teachers must balance linguistic expertise with an understanding of the subject matter and grading criteria. Given the open-ended nature of written responses, the risk of misinterpretation and grading bias is substantially higher compared to multiple-choice formats.

As a result, many Thai educators prefer multiple-choice examinations, which are quicker and easier to grade. This choice, however, comes at the expense of assessing higher-order skills. On average, teachers in Thailand manage classes of 100–400 students per subject while teaching 7–21 hours weekly and handling administrative, extracurricular, and pastoral responsibilities (Göksoy & Akdağ, 2014). Such workloads leave little room for the detailed evaluation of written work, leading to an overreliance on more easily graded exam types.

1.3 Limitations of Current Assessment Practices

Although multiple-choice exams are efficient, they fail to capture the depth of student reasoning, creativity, and expressive ability. Recent studies highlight that selected-response formats tend to assess surface-level understanding, whereas open-ended responses are better at revealing students' conceptual thinking and misconceptions (Bai et al., 2025; Morris et al., 2024). Written examinations, despite their higher grading workload, remain one of the most effective means of evaluating comprehensive understanding and critical thinking, especially when compared with highly constrained test formats (Emirtekin, 2025; Fuller & Bixby, 2024).

In recent years, efforts to explore automated grading with advanced AI systems particularly large language models (LLMs) have expanded rapidly. These include rubric-guided prompting, structured evaluation protocols, and hybrid workflows that combine peer evalua-

tion or partial automation for efficiency (Poličar et al., 2025; Xie et al., 2024). However, these approaches still face limitations arising from the complexities of natural language and the diversity of student expression. For example, while rule-based or keyword-oriented methods can detect explicit patterns, they struggle with semantic nuance, paraphrasing, indirect reasoning, and context-dependent meaning. Even LLM-based approaches may exhibit inconsistent scoring behavior or rely on shallow reasoning shortcuts, deviating from human graders' expectations (Emirtekin, 2025; Wu et al., 2024).

1.4 Role of Bloom's Taxonomy in Assessment

Bloom's Taxonomy provides a structured framework for categorizing cognitive skills, ranging from basic recall to higher-order reasoning and critical thinking. Despite its origins in mid-20th-century educational theory, the taxonomy remains highly relevant in modern assessment and AI-enhanced learning research (Huber & Niklaus, 2025; Stanny, 2016). In the context of this study, Bloom's Taxonomy plays two central roles.

First, it serves as a pedagogical foundation for interpreting the cognitive processes targeted by each exam question. This alignment ensures that grading criteria and model behavior correspond to intended learning outcomes—a principle emphasized in modern AI-pedagogical models (Hmoud & Ali, 2024; Kwan et al., 2025).

Second, Bloom's levels are used as a pre-screening mechanism within the AI-assisted grading framework. Expert teachers label each exam item according to Bloom's categories, and these labels guide the selection of models and reasoning strategies. Recent work has shown that different Bloom levels may benefit from different AI strategies: for instance, tasks at lower cognitive levels might be handled reliably with general-purpose LLM prompts, while more complex analytical or creative tasks may require specialized prompting or hybrid AI-rule-based reasoning (Luo et al., 2025; Scaria et al., 2024).

By integrating Bloom's Taxonomy into the system, our approach achieves both pedagogical validity and technical structure, bridging established educational theory with AI-driven assessment strategies.

1.5 Advances in Natural Language Processing (NLP)

Recent breakthroughs in Natural Language Processing (NLP) and the advent of Large Language Models (LLMs) have opened new possibilities for automated exam evaluation (Agarwal et al., 2021). LLMs demonstrate strong capabilities in tasks such as text classification, summarization, sentiment analysis, and automated evaluation across multiple languages. Unlike traditional rule-based methods, LLMs leverage deep learning and

attention mechanisms to capture contextual meaning, making them suitable for handling open-ended and nuanced responses.

For the Thai language, the application of LLMs introduces unique opportunities and challenges. While most existing LLMs are trained primarily on English or other widely spoken languages, recent developments have included multilingual models capable of handling Thai to varying degrees. By integrating LLMs with rule-based reasoning, it may be possible to design a hybrid evaluation framework that combines the flexibility and contextual understanding of LLMs with the consistency and transparency of structured rules.

1.6 Research Motivation and Contribution

The motivation behind this research is twofold. First, it seeks to reduce the grading burden on Thai teachers while enhancing fairness and consistency in evaluating written exam responses. Manual grading of open-ended answers often demands considerable time and effort, and is prone to subjectivity and variation across raters. By introducing an AI-assisted approach, the research aims to support teachers in achieving more objective, transparent, and efficient assessment practices. Second, it strives to promote the wider adoption of open-ended questions in Thai education, as these question types encourage students to demonstrate higher-order thinking skills such as critical reasoning, creativity, and written expression—competencies that are essential for lifelong learning but often underemphasized in traditional exam formats.

This thesis focuses on the development of an AI-assisted grading framework for Thai-language written exam questions. The framework integrates large language models (LLMs) with rule-based reasoning to combine the interpretive flexibility of AI with the explicit structure of human-designed rubrics. Through this hybrid approach, the system aims to improve grading accuracy, consistency, and efficiency while ensuring that its reasoning remains interpretable and reviewable by educators. The framework is designed not to replace teachers but to serve as an intelligent support tool that complements human judgment in the grading process.

The contributions of this research are multifaceted. It provides a comprehensive analysis of the challenges inherent in grading Thai written responses, identifying linguistic and pedagogical factors that affect evaluation consistency. It also presents a proof-of-concept framework that integrates LLMs and rule-based reasoning, offering a practical model for AI-assisted assessment in the Thai educational context. Furthermore, the study includes an empirical evaluation comparing the framework's performance with that of human graders in terms of fairness, consistency, and interpretability. Finally, the findings of this research

offer valuable insights into how AI-driven assessment tools can influence future educational assessment design, encourage the use of open-ended evaluations, and support curriculum development aimed at fostering analytical and expressive skills in students.

1.7 Research Objectives

The objectives of this research are as follows:

1. To explore the challenges and limitations in grading Thai written exam responses. This objective aims to investigate the underlying issues that make Thai-language written assessment difficult to evaluate consistently and fairly. It examines factors such as linguistic ambiguity, variations in student writing style, and the interpretive nature of open-ended questions. By understanding these challenges, the study seeks to identify the gaps where automation or AI assistance could meaningfully support human grading.
2. To build a proof-of-concept AI-assisted grading framework that integrates large language models (LLMs) with rule-based reasoning. The goal is to design and implement a hybrid framework that combines the contextual understanding of LLMs with explicit rule-based logic to improve reliability and transparency in the grading process. This framework serves as an experimental prototype demonstrating how AI can analyze, interpret, and score Thai written responses in a structured and explainable manner.
3. To compare the performance of the proposed framework with human grading in terms of fairness, consistency, and interpretability. This objective focuses on evaluating how closely the framework's scoring aligns with human teachers' judgments. The comparison emphasizes fairness—ensuring that the system treats responses objectively across scoring levels; consistency—maintaining stable grading patterns across different responses; and interpretability—providing clear reasoning that can be reviewed and understood by educators.
4. To demonstrate how AI can support, rather than replace, teachers in Thai educational contexts. This research highlights the collaborative potential of AI tools in education. Rather than automating the entire grading process, the framework is envisioned as an assistive system that provides preliminary evaluations, insights, and rubric-based guidance to teachers. It aims to show how AI can reduce grading workload, enhance efficiency, and promote fairness while keeping teachers at the center of the decision-making process.

1.8 Scope of the Study

This study is limited to the following scope:

1. The research focuses on student-level written exams under the Royal Society of Thailand, which serve as standardized Thai-language assessments. These exam questions are representative of authentic written evaluations used to assess comprehension, reasoning, and expression in the Thai context.
2. The study covers short-answer and essay-type questions evaluated using a 0–3 scoring scale. This scale provides a manageable yet informative range for distinguishing different levels of understanding and correctness in written responses.
3. The proposed framework employs selected large language models (LLMs) in combination with teacher-designed rubrics. The LLMs are responsible for generating initial assessments and explanations, while the rubrics provide structured criteria that guide both human and AI evaluation to ensure fairness and alignment with educational standards.
4. All exam questions are pre-labeled according to Bloom’s Taxonomy by expert teachers. These cognitive level labels are used to guide model selection and analysis, allowing the study to explore how the framework performs across different levels of cognitive demand.
5. The research focuses on evaluating the framework’s performance and analytical outcomes—particularly fairness, consistency, and interpretability—rather than implementing or deploying it in a live classroom environment. The study thus provides an experimental foundation and conceptual validation for potential future applications in Thai education.

1.9 Structure of the Thesis

The remainder of this thesis is organized as follows:

- **Chapter 2 – Literature Review:** Reviews existing work on exam evaluation, NLP in education, and applications of LLMs in AI-assisted grading frameworks.
- **Chapter 3 – Methodology:** Describes the proposed framework, including data collection, preprocessing, model selection, and rule-based integration.

- **Chapter 4 – Result and Discussion:** Presents the experimental findings, discusses the strengths and limitations of the framework, and compares it with baseline methods.
- **Chapter 5 – Conclusion and Future Work:** Summarizes key findings, highlights contributions, and suggests directions for future research.

CHAPTER 2

LITERATURE REVIEW

This chapter reviews prior work on automated exam evaluation, the use of large language models (LLMs) in education, and cognitive assessment frameworks. The thesis does not aim to build a new grading system but to test how reliable LLM-generated results are when compared with teacher evaluations. The review focuses on three areas: (1) applications of LLMs in educational assessment, (2) their reasoning and text-understanding abilities, and (3) approaches for validating AI results against human judgments. The chapter closes with Bloom’s taxonomy as a framework for aligning both AI and teacher assessments.

2.1 The Integration of LLMs in Educational Assessment

2.1.1 State of the art and practice in AI in education

As highlighted by Holmes and Tuomi (2022), the accelerating advancement of Large Language Models (LLMs) has intensified interest in their potential to automate educational processes—particularly feedback and assessment. Early studies demonstrated that LLMs can analyze written responses and provide formative feedback with a level of fluency and contextual understanding previously unattainable by rule-based systems. The integration of artificial intelligence (AI) into education, or Artificial Intelligence in Education (AIED), can be categorized into three major domains: student-focused, teacher-focused, and institution-focused applications. Each category serves distinct purposes but shares the goal of improving learning outcomes and administrative efficiency.

Student-focused applications include intelligent tutoring systems, adaptive learning platforms, and personalized feedback mechanisms that respond dynamically to each learner’s needs. Teacher-focused tools encompass automated grading systems, plagiarism detection, and analytics dashboards that help educators identify student misconceptions or engagement levels. Institution-focused systems, meanwhile, support large-scale educational management, such as admissions decision-making, e-proctoring, and curriculum analytics. These technological developments promise to enhance scalability, ensure consistency in grading, and provide timely feedback—particularly valuable in contexts where human resources are limited.

Despite these advantages, several critical challenges accompany AI adoption in ed-

ucation. Key concerns include cultural and linguistic bias, fairness, ethical transparency, and the lack of robust empirical evidence validating actual educational impact. Many AI-driven assessment tools remain test-centric, often neglecting broader educational goals such as socio-emotional development, creativity, and critical reasoning. This limitation is especially pronounced in underrepresented languages and cultural contexts.

For the Thai educational landscape, these concerns are even more pronounced. Compared to high-resource languages like English or Chinese, Thai suffers from limited data availability, linguistic diversity across regions, and nuanced expressions that challenge direct model generalization. As a result, LLM-based systems trained primarily on non-Thai data risk perpetuating biases and inconsistencies in grading. Such biases can manifest not only in content understanding but also in scoring fairness, potentially disadvantaging students who express themselves in culturally or linguistically distinct ways.

The present thesis directly addresses this gap by developing and evaluating an AI-assisted grading framework tailored for Thai-language written examinations. By combining LLM-based scoring with rule-based reasoning and human-designed rubrics, the framework aims to achieve both consistency and fairness. Human teachers play a crucial role in refining rubrics and validating model performance, ensuring that the automated system remains transparent and pedagogically aligned. In doing so, this research contributes to the broader discourse on equitable AI in education—demonstrating that effective integration of LLMs in assessment requires not only technical sophistication but also cultural and linguistic sensitivity.

2.1.2 Teachers’ agency in the era of LLM and generative AI: Designing pedagogical AI agents

Building upon this perspective, Lan and Chen (2024) highlights the dual nature of LLMs in educational contexts—acting both as a transformative tool for personalized learning and as a potential source of academic integrity challenges. On the positive side, LLMs can deliver tailored explanations, adaptive feedback, and individualized support aligned with each learner’s pace and style. However, the same capabilities that enable personalization can also encourage academic dishonesty, as students may become overly reliant on AI-generated outputs rather than engaging in independent cognitive effort. This dichotomy underscores the necessity for thoughtful pedagogical design and ethical frameworks governing how LLMs are integrated into learning environments.

In particular, studies examining teacher and student perceptions of generative AI

reveal key challenges. Educators report difficulty distinguishing between student- and AI-generated work, uncertainty about AI's pedagogical value, and concerns that excessive dependence could undermine authentic learning. Students, meanwhile, often adopt AI tools superficially, using them primarily as search engines or text generators rather than as interactive learning companions that foster reflective thinking, critical reasoning, or metacognitive development.

To address these issues, the concept of Pedagogical AI Agents has been proposed—AI-driven teaching assistants that collaborate with human teachers within a team-teaching model. In this framework, the teacher remains central, responsible for instructional planning, rubric design, and student evaluation, while the AI agent delivers adaptive content, real-time feedback, and continuous learning analytics. Structured around established instructional design models such as ADDIE and Dick & Carey, the approach emphasizes role definition, instructional procedures, assessment rubrics, guidance and scaffolding, and feedback mechanisms. For example, AI agents have been employed to teach sequencing words in English, providing structured explanations, contextual examples, and interactive exercises with immediate feedback, leading to measurable improvements in understanding and retention.

Ultimately, this line of work reinforces that AI should complement rather than replace teachers, acting as a collaborative co-teacher that extends educators' reach and efficiency while preserving fairness and pedagogical oversight. Within the context of the present thesis, this perspective supports positioning AI as a partner in Thai-language written exam evaluation—ensuring that automation enhances scoring consistency and efficiency without compromising human judgment, contextual understanding, or ethical responsibility.

2.1.3 Exam in the time of chatgpt

As discussed in Ryznar (2023), recent research examines the implications of LLMs—specifically ChatGPT—on academic integrity and assessment design in higher education. The extraordinary capabilities of ChatGPT—such as its strong performance in complex professional or legal examinations—raise fundamental questions about the validity of conventional assessment methods. As LLMs become increasingly adept at generating coherent, domain-specific responses, educators face the risk that student submissions may no longer reflect genuine individual understanding or reasoning ability.

Several critical concerns are identified. Students may over-rely on AI tools to complete assignments or exam tasks, undermining independent thinking. Teachers often struggle to distinguish AI-generated from human-authored responses, as existing plagiarism detection tools are limited. Additionally, online examination systems often lack adequate safe-

guards, particularly in remote contexts, threatening fairness and reliability in academic evaluation.

To mitigate these risks, multiple strategies are suggested, including video-based proctoring, lockdown browsers, AI-specific plagiarism detection, and strict time limits. However, technical controls alone are insufficient. Effective exam design—emphasizing problem-solving, issue-spotting, logical reasoning, and analytical synthesis—is recommended as the most sustainable and pedagogically sound approach. Illustrative examples from legal education show that while ChatGPT performs well on definition-based questions, it struggles with analytical or case-based scenarios, highlighting the importance of measuring higher-order cognitive skills rather than mere factual recall.

The study concludes that eliminating AI from education is neither feasible nor desirable. Instead, institutions must adapt assessment practices to coexist with AI responsibly. A multi-layered approach that combines sound exam design, fairness-oriented rubrics, and procedural safeguards can preserve both integrity and authenticity.

This perspective aligns directly with the present thesis, which emphasizes structured rubric-based evaluation and human oversight as mechanisms to ensure fairness and validity in AI-assisted grading. By incorporating clear teacher-designed rubrics, cross-verification between AI and human assessors, and transparent reasoning processes, the proposed framework addresses the challenges posed by generative AI. In the Thai educational context—where linguistic diversity and limited AI training data further complicate fairness—well-designed assessment structures become particularly essential, reinforcing the thesis’s central contribution: an LLM-based framework that is pedagogically sound, culturally fair, and empirically reliable.

2.1.4 Summary

Taken together, these studies illuminate both the transformative potential and the inherent complexities of integrating LLMs into educational assessment. Collectively, they demonstrate that while AI technologies can enhance scalability, efficiency, and consistency in grading, their deployment must be guided by principles of fairness, transparency, and pedagogical integrity. Holmes and Tuomi (2022) establish the foundational perspective that AI can democratize access to feedback and streamline educational processes, yet they caution that issues of cultural and linguistic bias remain unresolved—particularly in underrepresented languages such as Thai. Lan and Chen (2024) further extend this discussion by emphasizing the need for human–AI collaboration in the form of pedagogical AI agents, which complement rather than replace teachers. Their team-teaching model reinforces the

idea that successful educational AI systems depend on clearly defined human roles, robust rubrics, and ethical instructional design.

In contrast, Ryznar (2023) highlights the other side of AI integration—the risks associated with academic dishonesty, overreliance, and declining authenticity in assessment. His findings reveal that without thoughtful redesign, conventional exam formats are vulnerable to misuse and may fail to capture genuine student learning. Consequently, future assessment practices must evolve beyond mechanical grading to emphasize reasoning, creativity, and critical thinking—dimensions that LLMs can support but not fully replicate.

Across these perspectives, a unifying theme emerges: the future of AI-assisted assessment lies in balance. Automation should serve as a tool that augments human judgment, not a mechanism that replaces it. Effective integration of LLMs therefore requires systems that are not only technically capable but also contextually grounded—sensitive to linguistic, cultural, and pedagogical nuances within each educational environment.

In the context of Thai-language education, where limited data resources and unique linguistic structures present additional challenges, these insights are particularly pertinent. The thesis builds upon this foundation by proposing an AI-assisted grading framework that integrates LLM-generated evaluations with teacher-designed rubrics and expert verification. This approach directly addresses the concerns raised in prior research—offering a pathway toward AI assessment that is efficient yet fair, data-driven yet human-centered, and adaptable to the cultural realities of Thai education. These opportunities and challenges are summarized in Figure 2.1.

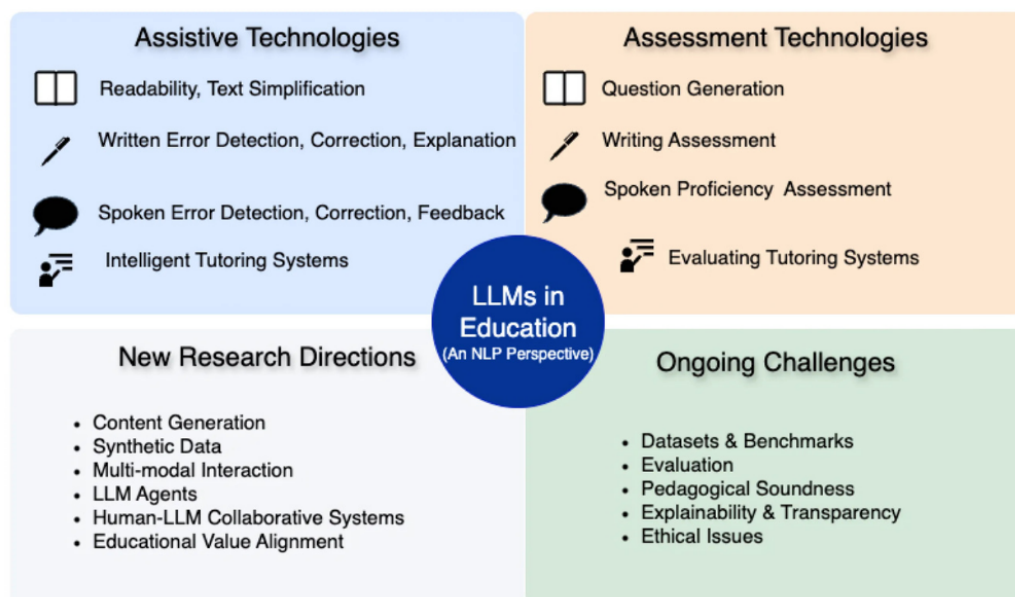


Figure 2.1 Opportunities and Challenges of LLMs in Educational Assessment. Adapted from Vajjala et al. (2025).

2.2 Leveraging LLMs for Text Understanding and Cognitive Inference

2.2.1 Consistency Analysis of ChatGPT

As shown in Jang and Lukasiewicz (2023), while LLMs are widely recognized for their impressive text generation capabilities, their reliability in logical reasoning and semantic inference remains inconsistent. Although GPT-4 demonstrates strong performance in syntactic and linguistic comprehension tasks, it continues to struggle with more complex reasoning forms such as negation, contradiction, and inference-based decision-making. These findings suggest that while LLMs excel at capturing surface-level language patterns, their ability to represent deeper cognitive reasoning—particularly those involving relational and causal logic—remains limited.

A more systematic investigation of this limitation is presented in the Consistency Analysis of ChatGPT (2023), which evaluates the trustworthiness and logical consistency of ChatGPT and GPT-4 through four categories of reasoning tasks: (1) semantic reasoning, which assesses the model’s understanding of meaning and contextual relationships; (2) negation reasoning, which tests comprehension of oppositional statements; (3) symmetric reasoning, where equivalence between reversed elements (e.g., “A relates to B” versus “B relates to A”) is measured; and (4) transitive reasoning, which examines the model’s ability to infer indirect relationships (e.g., if $A \rightarrow B$ and $B \rightarrow C$, then $A \rightarrow C$).

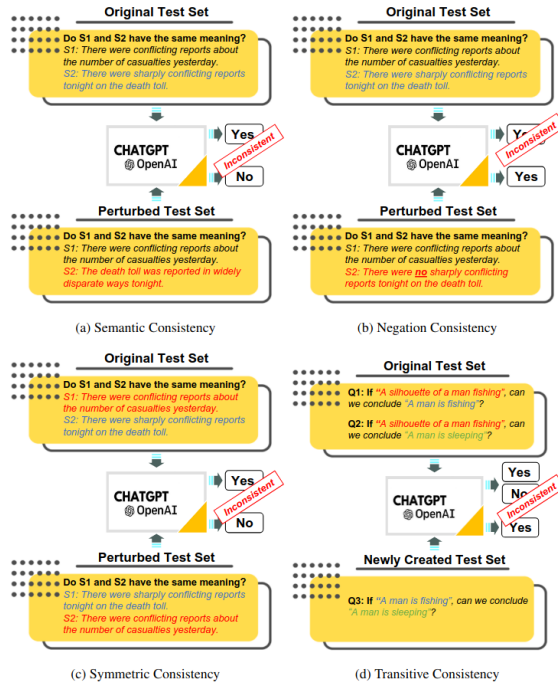


Figure 2.2 Consistency evaluation process of (a) semantic, (b) negation, and (c) symmetric consistency on MRPC and (d) transitive consistency on SNLI, adapted from Jang and Lukasiewicz (2023).

The study revealed that, while GPT-4 and ChatGPT have shown measurable improvement in handling negation and transitive inference compared to earlier LLM generations, they still fall short in maintaining semantic and symmetric consistency. Notably, both models exhibited self-contradiction when presented with paraphrased inputs or minor variations in question order, exposing an inherent input-order sensitivity that undermines reliability. These inconsistencies persisted even when using larger model architectures, advanced prompting strategies, or few-shot learning techniques—suggesting that scaling alone does not resolve fundamental reasoning flaws.

Furthermore, the authors caution that continual model enlargement is not a sustainable solution, given its escalating economic cost and environmental impact. Instead, they argue that the future of reliable AI reasoning depends on introducing structured, rule-based, and human-guided approaches that constrain model outputs within logically coherent boundaries. Logical consistency, they contend, is a prerequisite for deploying LLMs in high-stakes decision-making domains such as law, healthcare, and finance—fields where even subtle contradictions can result in serious consequences.

In the context of educational assessment, these insights carry critical implications. Logical reliability is central to the fairness and transparency of automated grading. If an LLM inconsistently interprets similar student responses or fails to apply reasoning consistently across equivalent contexts, the resulting evaluation cannot be considered trustworthy. The present thesis directly addresses this concern by integrating rule-based reasoning with LLM evaluation and teacher-validated rubrics, ensuring that AI-driven assessments remain logically sound and contextually fair. By combining structured rubrics and human oversight, the proposed framework mitigates the weaknesses identified in existing LLM reasoning research—offering a model that aligns machine inference with educational judgment in a transparent, reproducible manner.

2.2.2 Enhanced Sentiment Intensity Regression Through LoRA Fine-Tuning on Llama 3

As demonstrated in Lin et al. (2024), beyond reasoning consistency, another research direction explores the alignment of LLMs with domain-specific cognitive tasks through efficient fine-tuning techniques. The study investigated Enhanced Sentiment Intensity Regression Through LoRA Fine-Tuning on LLaMA-3, aiming to improve the model's sensitivity to emotional gradation rather than mere polarity classification. Unlike conventional sentiment analysis, which focuses on categorical outputs (positive, negative, or neutral), this

work targets the continuous prediction of emotional strength, providing a more nuanced understanding of linguistic affect.

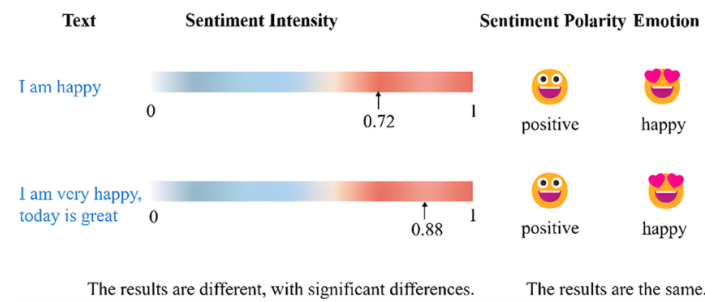


Figure 2.3 Comparison of sentiment intensity regression with other sentiment analysis tasks. The intensity regression approach highlights differences between example sentences, whereas classification and emotion detection yield identical categorical outputs. Adapted from Lin et al. (2024).

To achieve this, Lin and colleagues first enhanced the RoBERTa model by integrating an Adaptive Weighted Huber Loss and an Additive Attention Mechanism, both designed to improve prediction precision and contextual focus. The refined outputs from this model were then leveraged to fine-tune LLaMA-3 using Low-Rank Adaptation (LoRA)—a parameter-efficient technique that enables rapid specialization without the computational cost of full retraining. Complementary methods, including prompt engineering and data augmentation, further strengthened the model’s robustness and generalization.

Experimental results on the SemEval 2017 and 2018 benchmarks demonstrated that the LoRA-fine-tuned model outperformed several baselines—such as XLNet, ALBERT, Gemma-7B, and the standard LLaMA-3—in both error reduction and correlation with human judgments. Ablation studies confirmed that both the custom loss function and the attention mechanism contributed critically to these improvements, while text augmentation enhanced resilience to linguistic variation. Despite these gains, the authors acknowledged the limitations of dataset size and computational scale, noting that broader testing with larger models and multimodal inputs remains an important future direction.

This work underscores a key insight: LLMs cannot excel universally across all cognitive dimensions without task-aligned adaptation. Rather than treating a model as a one-size-fits-all evaluator, fine-tuning allows for the calibration of linguistic sensitivity, reasoning focus, and contextual awareness. In the context of this thesis, this finding reinforces the rationale for designing a structured, rubric-guided, and rule-based evaluation framework, where model outputs are aligned with teacher-defined cognitive dimensions. Such an approach parallels fine-tuning in that it systematically constrains and guides the model’s interpretive

space—enhancing both the reliability and the educational validity of AI-assisted grading.

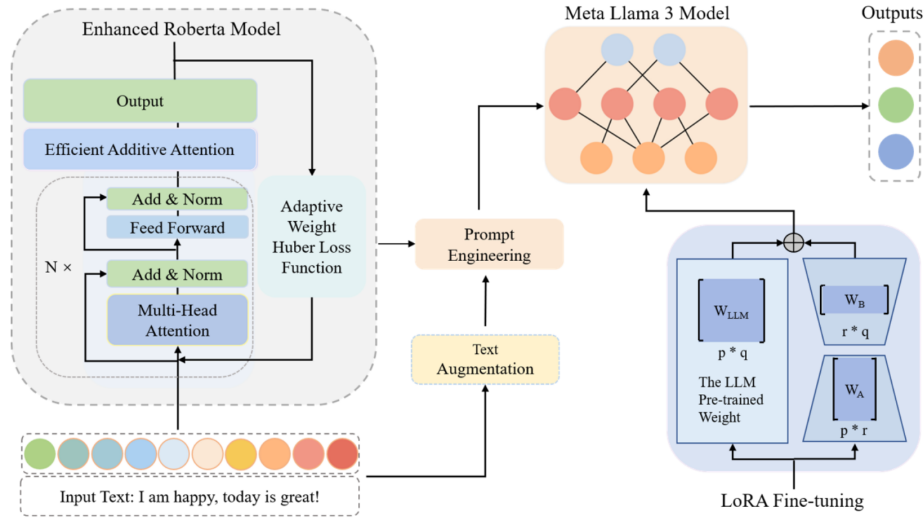


Figure 2.4 Workflow of the Enhanced Sentiment Intensity Regression Model: Input text is processed by the enhanced Roberta model with efficient additive attention and adaptive weighted Huber loss. The output is combined with augmented input data for prompt engineering, followed by fine-tuning of the Llama 3 model using LoRA within the Unsloth framework. Adapted from Lin et al. (2024).

2.2.3 Summary

Taken together, these studies reveal that while LLMs have made substantial progress in linguistic comprehension and contextual reasoning, their reliability remains uneven across different types of inference and emotional interpretation. Logical consistency studies underscore persistent vulnerabilities in semantic coherence, symmetry, and contradiction handling, while fine-tuning research highlights the potential of task-specific adaptation to enhance precision in cognitive and affective understanding.

Building on these insights, this thesis adopts Bloom’s taxonomy as a guiding cognitive framework to classify and evaluate student responses according to their reasoning depth and expressive complexity. By mapping answers to distinct cognitive levels, the system determines when to employ rule-based reasoning (for factual, definitional, or procedural tasks) and when to leverage LLM-based evaluation (for interpretive or analytical responses). This hybrid design mitigates the “one-size-fits-all” limitation of general-purpose models, ensuring that the assessment strategy aligns with both the intended learning outcomes and the cognitive capabilities of the AI.

Ultimately, this approach bridges the gap between linguistic pattern recognition and meaningful reasoning—moving toward AI-assisted grading that is not only automated and efficient but also cognitively grounded and pedagogically faithful.

2.3 LLM Ensembles and Model Selection Strategies

2.3.1 QUAD-LLM-MLTC: Large Language Models Ensemble Learning for Healthcare Text Multi-Label Classification

As demonstrated in Sakai and Lam (2025), no single LLM excels universally across all tasks, leading ensemble-based strategies to emerge as a promising direction to mitigate individual model weaknesses and enhance reliability. The study introduced QUAD-LLM-MLTC, an ensemble framework designed for healthcare multi-label text classification, where semantic ambiguity and data imbalance present significant challenges. The framework integrates four distinct LLMs—BERT for semantic token extraction, PEGASUS for text augmentation, GPT-4o for classification, and BART for probabilistic label estimation—within a coordinated pipeline.

By combining outputs through a meta-classifier, the system achieved substantial gains in F1-score and robustness compared to both traditional machine learning approaches and single-model baselines. Notably, the method operated under zero-shot conditions, demonstrating that careful orchestration of multiple LLMs can yield generalizable and contextually aware results even without large annotated datasets. This architecture highlights the value of contextual prompt engineering, label probability fusion, and multimodel design, which collectively enhance semantic precision and consistency across tasks.

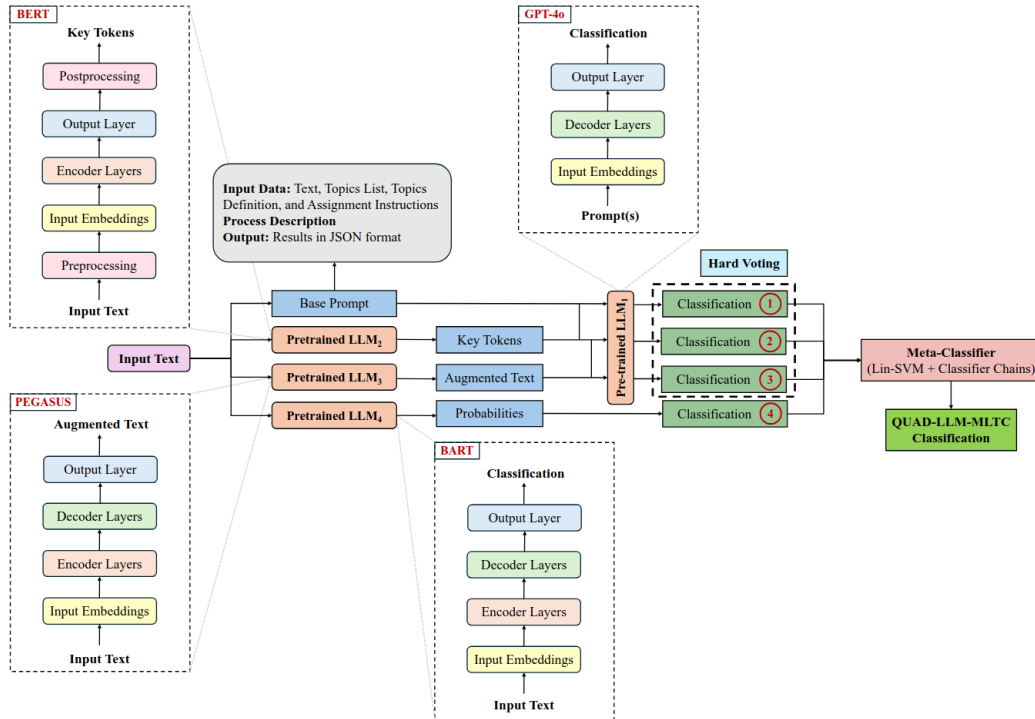


Figure 2.5 Overview of the QUAD-LLM-MLTC architecture, illustrating the four-model pipeline for multi-label classification. Adapted from Sakai and Lam (2025).

The findings from this work underscore a critical insight also central to this thesis: reliability and fairness in AI-driven assessment depend not solely on model size or capability but on structural design and cooperative reasoning. Just as QUAD-LLM-MLTC leverages multiple specialized models to balance strengths and weaknesses, the proposed AI-assisted grading framework in this study similarly integrates rule-based reasoning, rubric-guided evaluation, and teacher validation. This hybrid structure ensures that automated grading remains not only efficient and scalable but also transparent, interpretable, and pedagogically aligned with human judgment.

2.3.2 LLM-TOPLA: Efficient LLM Ensemble by Maximising Diversity

As demonstrated in Tekin et al. (2024), LLM-TOPLA is a diversity-optimized ensemble framework designed to enhance both robustness and efficiency across a variety of reasoning and generation tasks. Unlike conventional ensemble methods that prioritize averaging model predictions, LLM-TOPLA explicitly measures and leverages error diversity among base models to form more balanced and resilient outcomes. Its architecture integrates three core innovations: a Focal Diversity Metric for quantifying complementary model behavior, a Diversity-Optimized Ensemble Pruning Algorithm for selecting high-performing sub-ensembles, and a Learn-to-Ensemble mechanism that adapts output fusion strategies to task type—TOPLA-Weighted for reasoning tasks and TOPLA-Summary for generative ones.

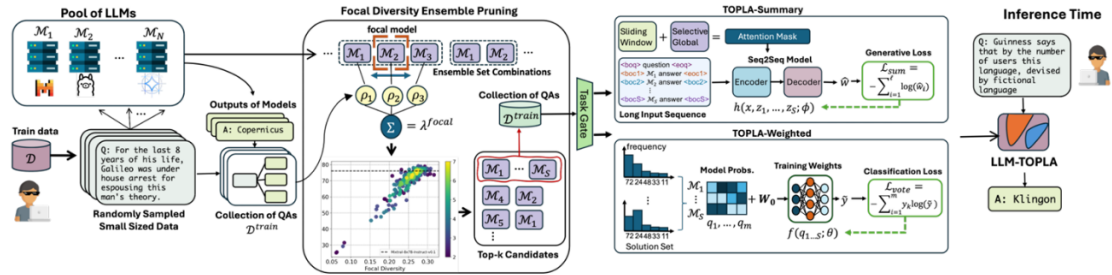


Figure 2.6 Overview of the LLM-TOPLA framework, illustrating diversity-based ensemble selection and adaptive fusion mechanisms. Adapted from Tekin et al. (2024).

Evaluated across six benchmark datasets—MMLU, GSM8k, SearchQA, XSum, BBH, and ARC—the framework achieved consistent performance gains over leading ensemble methods such as Mixtral, More Agents, and LLM-Blender. Results showed 2–8% accuracy improvements in reasoning benchmarks and over 30% increases in F1 and ROUGE scores for generative tasks, all while maintaining high efficiency with fewer sub-models. This demonstrates that fostering model diversity and structured output fusion can outper-

form even large-scale fine-tuning approaches, offering a scalable path toward reliable LLM orchestration.

The insights from LLM-TOPLA directly parallel the hybrid philosophy of this thesis. Both emphasize that structured diversity—whether through model ensembles or rubric-guided reasoning—serves as the foundation of fairness and stability in AI-driven evaluation. In the proposed grading framework, this principle manifests through the integration of rule-based reasoning, rubric alignment, and teacher moderation, ensuring that no single evaluation component dominates. By balancing automated inference with human-informed structure, the system aspires to deliver assessments that are robust, interpretable, and pedagogically consistent across cognitive levels and linguistic variations.

2.3.3 Summary

These studies collectively demonstrate that strategic model selection and structured ensemble design can substantially improve the reliability, robustness, and interpretability of LLM-based systems. By combining models with complementary strengths or integrating diversity-aware mechanisms, researchers have achieved more stable and context-sensitive outcomes across domains.

Building on this insight, the present study employs a hybrid evaluation strategy: selecting the best-performing LLM for each Bloom’s taxonomy category while incorporating rule-based reasoning for tasks that require deterministic and explainable assessment. This approach balances the adaptability of language models with the rigor of rule-based logic, ensuring evaluations that are not only accurate and fair, but also transparent and pedagogically grounded—a crucial requirement for Thai-language educational contexts.

2.4 Bloom’s Taxonomy in Educational AI and Cognitive Classification

Bloom’s taxonomy provides a structured theoretical foundation for aligning assessment with cognitive skill levels. It divides learning objectives into six hierarchical categories, each representing a progressive stage of cognitive development. The taxonomy not only guides curriculum design and learning objectives but also serves as a framework for evaluating the depth of understanding demonstrated in student responses—making it particularly useful for AI-supported assessment systems.

1. **Remembering** – recalling factual knowledge and basic concepts. This foundational level involves recognizing, identifying, and retrieving information from memory. Assessments targeting this level typically include direct recall or multiple-choice questions that measure knowledge retention rather than comprehension.

2. **Understanding** – explaining ideas or concepts. Learners at this level demonstrate comprehension by interpreting, summarizing, or classifying information in their own words. Evaluation focuses on whether students can grasp meaning and translate it across contexts rather than merely restating facts.
3. **Applying** – using knowledge in new contexts. This level reflects the ability to transfer theoretical knowledge to practical or situational problems. Students apply learned concepts to execute procedures, solve problems, or demonstrate techniques, making it a key stage for performance-based assessment.
4. **Analyzing** – breaking down information into components and recognizing relationships or patterns. Analytical thinking involves distinguishing between evidence and argument, identifying cause-and-effect relationships, and examining underlying assumptions. This level assesses critical reasoning and logical structure in student responses.
5. **Evaluating** – making judgments or justifying decisions based on criteria or standards. Learners are expected to critique, defend, or assess ideas using well-defined criteria. At this stage, assessment measures reasoning quality, justification of opinions, and the ability to evaluate alternatives—key for rubric-based evaluation systems.
6. **Creating** – producing new or original work by integrating knowledge and creativity. The highest cognitive level involves synthesizing ideas to form new solutions, models, or perspectives. This level tests originality, innovation, and the ability to generate coherent, meaningful output—making it a strong indicator of higher-order thinking.

Each level in Bloom’s taxonomy reflects increasing cognitive complexity, from surface-level recall to deep conceptual reasoning and creativity. When applied to automated grading, these categories allow AI systems to align evaluation criteria with learning objectives, ensuring that the assessment process captures both the *depth* and *quality* of student understanding. This hierarchical structure is especially valuable in this thesis, serving as the cognitive backbone for mapping responses to appropriate evaluation strategies—rule-based for lower levels and LLM-assisted for higher-order reasoning.

2.4.1 Exam questions classification based on Bloom’s taxonomy cognitive level using classifiers combination

As demonstrated in Abduljabbar and Omar (2015), Bloom’s taxonomy has been widely adopted in educational AI research as a framework for classifying both questions

and student responses according to cognitive complexity. Their work showed the effectiveness of using machine learning to automatically categorize questions into Bloom's levels, highlighting its value in structuring automated evaluation systems.

Table 2.1 Six Categories of the Cognitive Domain Based on the Revised Bloom's Taxonomy (2001) With Illustrative Examples

Category	Sample Keywords	Sample Behaviours	Sample Questions
Remembering	Define, list, name, recall, recognize	Students are able to define the 6 levels of Bloom's taxonomy	Recall basic facts and terms.
Understanding	Classify, explain, describe, discuss, explain, summarize.	Students can explain the purpose of Bloom's taxonomy.	Explain relationships among concepts.
Applying	Demonstrate, implement, solve, use, execute	Students write an instructional objective for a method in the program.	Apply knowledge to new situations
Analyzing	Analyze, compare, contrast, differentiate, examine.	Students compare and contrast cognitive and affective domains.	Break concepts into parts and examine relationships.
Evaluating	Assess, critique, defend, judge, validate.	Students judge the effectiveness of Revised Bloom's taxonomy.	Justify the concept of inheritance and give code examples.
Creating	Assemble, construct, design, formulate, invent.	Students design a classification method for educational objectives.	Combine elements to form a coherent whole or original product.

Recent work on automatic exam question classification further advances this line of research by addressing the inherent difficulty of mapping short and sparse textual data to cognitive levels. Because exam questions are typically concise and context-limited, traditional classification methods often suffer from low accuracy and poor generalization.

To overcome these challenges, researchers have proposed a Voting Ensemble Model that combines the strengths of multiple classifiers—namely Support Vector Machine (SVM), Naïve Bayes (NB), and k-Nearest Neighbour (k-NN). In addition, feature selection techniques such as Chi-Square, Mutual Information, and Odds Ratio were employed to isolate the most informative features. Experimental results showed that the ensemble approach outperformed all individual models, with the best accuracy achieved using Mutual Information

and 250 selected features.

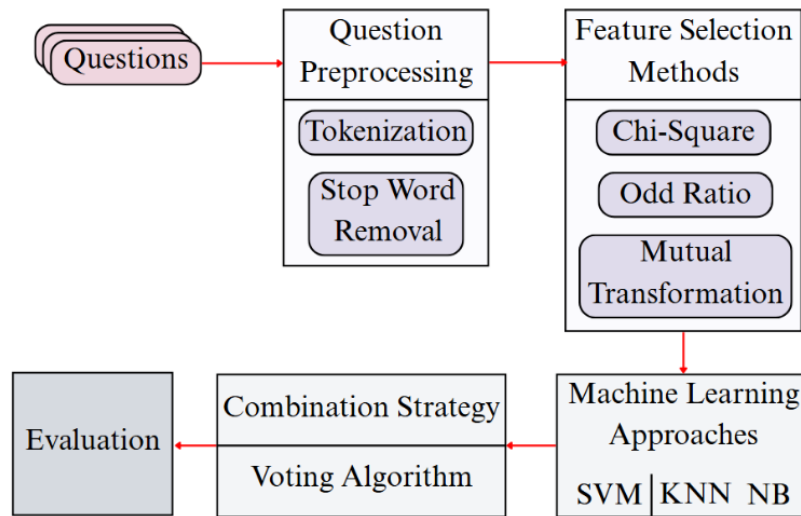


Figure 2.7 Overview of the proposed classifier combination model for Bloom’s-level prediction. Adapted from Abduljabbar and Omar (2015).

These findings highlight that ensemble learning enhances the accuracy, stability, and scalability of Bloom’s-level classification, particularly in handling short-text educational data. This approach provides a robust foundation for fine-grained exam analysis and adaptive assessment design.

From the perspective of this thesis, such work directly supports the integration of Bloom’s taxonomy as the structural backbone of the proposed AI-assisted grading framework. It also reinforces the rationale for employing machine learning and ensemble-based strategies to achieve more precise and cognitively aligned evaluation of exam content.

2.4.2 On Evaluating the Contribution of Validation for Question Answering

Rodrigo and Penas (2015) emphasized the importance of inter-rater agreement and gold-standard datasets, both critical for training reliable assessment systems. Building on this, subsequent research has examined how validation processes influence the development and reliability of Question Answering (QA) systems, especially in the context of answer validation (AV) modules that verify the correctness of responses.

Traditional evaluation metrics such as Accuracy, Precision, Recall, and F1-score often fail to reflect the real-world effectiveness of AV modules. These intrinsic measures assess a module’s internal consistency but do not indicate whether it actually improves the overall performance of the QA system.

To address this limitation, researchers proposed a QA-based evaluation framework that assesses AV modules based on their extrinsic impact—that is, their contribution to the

end-to-end performance of the full QA system. The framework introduces task-aligned metrics such as Mean Reciprocal Cost (MRC) and Macro-Precision, evaluated using the AVE 2008 dataset across both simulated and real QA systems.

Experimental results revealed that conventional evaluation methods could produce misleading conclusions in up to 88% of cases, whereas the proposed QA-based framework provided a more accurate reflection of true system performance. This evidence strongly supports the view that AV modules should not be assessed in isolation but as integrated components within the broader QA pipeline.

The study therefore argues for a shift from intrinsic to extrinsic evaluation, emphasizing that effective system validation must consider context, fairness, and real-world impact rather than raw numerical performance alone.

This perspective aligns closely with the approach adopted in this thesis. The proposed AI-assisted grading framework emphasizes context-aware evaluation that reflects genuine assessment outcomes and fairness. By incorporating structured rubrics, rule-based reasoning, and teacher validation, the system ensures that LLM evaluations of Thai-language exam responses are not only accurate but also transparent, interpretable, and pedagogically meaningful.

2.4.3 Machine Learning-based Automated System for Subjective Answer Evaluation

As demonstrated in Dodia et al. (2023), more recent studies explore hybrid approaches that combine rule-based methods with machine learning techniques to grade short-answer responses, aiming to reduce the time, cost, and workload associated with manual grading.

The proposed framework integrates three key components:

1. BERT embeddings – to capture the semantic meaning of both teacher and student answers;
2. Cosine similarity – to measure semantic closeness between vectorized responses; and
3. Jaccard distance – to assess lexical overlap between word sets in answers.

The similarity scores from these three techniques were weighted (3.5 : 1 : 0.5) and passed into a neural network with a sigmoid activation function to produce the final student score.

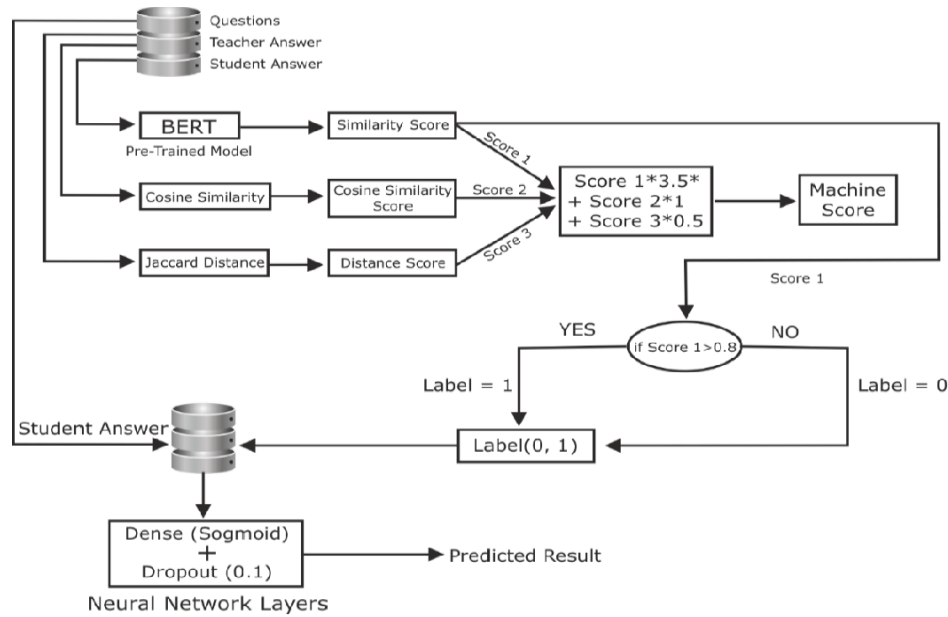


Figure 2.8 Schematic architecture of the proposed automated grading system, integrating BERT embeddings, cosine similarity, and Jaccard distance. Adapted from Dodia et al. (2023).

The dataset consisted of 100 exam questions, each paired with a model answer provided by a teacher and five student responses per question. Experimental results demonstrated that combining multiple similarity metrics achieved the best performance, with an accuracy of 91%, precision of 83%, recall of 79%, and an error rate of 9.01%. Compared to baseline methods such as LSTM, Naïve Bayes, and KNN, this approach produced scores that more closely aligned with teacher evaluations.

Overall, the study showed that combining semantic and statistical similarity measures within a neural network framework can significantly improve the reliability and accuracy of automated grading for descriptive answers.

This work directly supports the direction of the present thesis, illustrating that integrating semantic models with rule-based or similarity-based scoring methods enhances grading accuracy, fairness, and interpretability—key principles underpinning the design of the AI-assisted Thai exam evaluation framework.

2.4.4 How Question Types Reveal Student Thinking: An Experimental Comparison of Multiple-True-False and Free-Response Formats

While Hubbard et al. (2017) demonstrated that question formats (e.g., multiple-true-false vs. free-response) influence the depth of student reasoning captured. This study investigated how different exam formats—particularly Multiple-True-False (MTF) and Free-Response (FR)—affect students' conceptual understanding and reasoning processes in un-

dergraduate biology courses.

Using a crossover experimental design with over 400 students (Figure 2.9), the researchers compared performance, conceptual understanding, and diagnostic value across the two question types. Results showed a correlation in overall performance between MTF and FR formats, though students generally scored higher on MTF items.

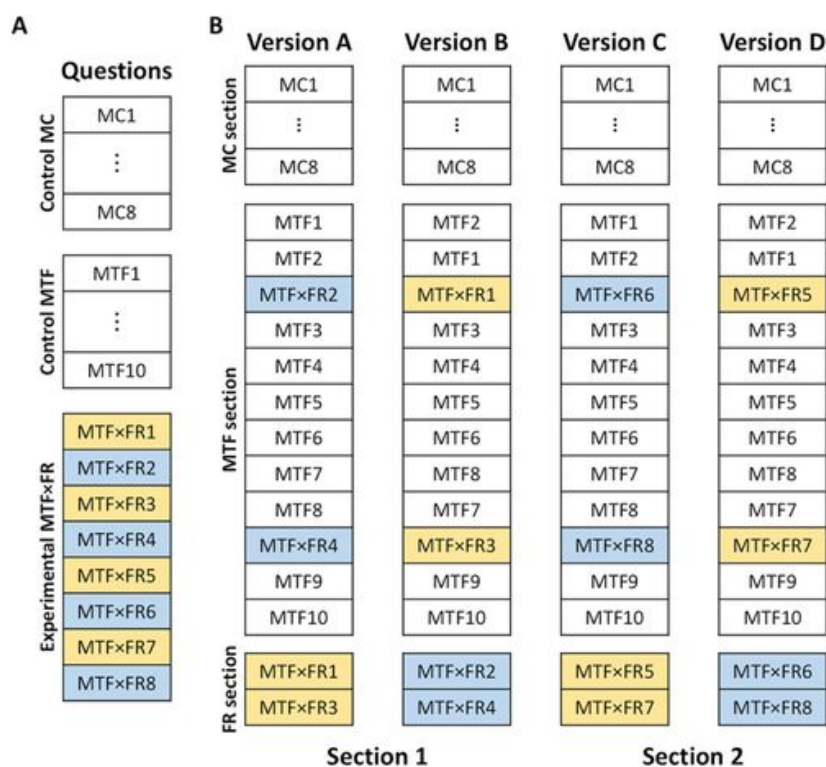


Figure 2.9 Crossover experimental design comparing MTF and FR question formats in undergraduate biology exams. Adapted from Hubbard et al. (2017).

MTF questions revealed mixed conceptions—students expressing both correct and incorrect ideas simultaneously—while FR responses contained fewer misconceptions but tended to be ambiguous or vague, particularly among lower-performing students.

In terms of scoring, partial-credit MTF formats tended to inflate scores, while all-or-nothing MTF yielded results more consistent with FR scoring.

Overall, the study found that MTF questions are effective for quickly identifying specific misconceptions, whereas FR questions better capture deeper cognitive reasoning, albeit with greater scoring complexity and subjectivity. The authors concluded that hybrid exam formats can provide a practical balance between diagnostic precision, grading efficiency, and authentic reflection of student understanding.

This finding aligns closely with the present thesis, emphasizing that exam design directly shapes assessment validity and interpretability. In the context of AI-assisted grading,

this underscores the need for structured rubrics and teacher validation to ensure that automated scoring of open-ended responses maintains both efficiency and fairness. Conceptually, MTF mirrors AI-based scoring—fast and consistent yet prone to bias, while FR parallels human grading—deeper but subjective. The proposed hybrid framework in this thesis—LLM-based scoring combined with rubric reasoning and teacher validation—therefore represents an ideal synthesis of both approaches, achieving balanced, interpretable, and trustworthy evaluation outcomes.

2.4.5 Summary

Collectively, these studies demonstrate that Bloom’s taxonomy provides not only a theoretical scaffold for understanding cognitive complexity but also a practical foundation for designing and validating AI-assisted assessment systems. By classifying exam questions and student responses along cognitive dimensions, researchers have shown improvements in grading accuracy, fairness, and interpretability. Ensemble and hybrid machine learning models further enhance this process, effectively addressing challenges in short or ambiguous responses through semantic and rule-based integration. Moreover, studies on question formats highlight that assessment design itself influences the depth and validity of reasoning captured, underscoring the importance of aligning question structure with evaluation strategy.

Building on these insights, the present study applies Bloom’s taxonomy as a classification and decision framework—mapping responses to appropriate cognitive levels and selecting the most suitable evaluation method, whether LLM-based, rule-based, or teacher-validated. This alignment ensures that AI-assisted grading achieves not only computational efficiency but also transparency, pedagogical relevance, and cognitive fidelity—key requirements for trustworthy educational assessment in Thai-language contexts.

2.5 Summary of Literature Review

The reviewed literature collectively reveals four major insights that form the conceptual foundation of this thesis:

- **AI and LLMs in Education:** Large Language Models (LLMs) have transformed educational technology by enabling scalable, personalized feedback and automated grading. However, scholars emphasize persistent challenges related to fairness, transparency, linguistic bias, and academic integrity, underscoring the need for teacher-centered and ethically guided integration.

- **Assessment Design and Validation:** Studies highlight that evaluation systems must go beyond technical accuracy to measure real-world effectiveness. Reliable assessment requires validation through inter-rater agreement, rubric alignment, and extrinsic performance metrics—ensuring that automated grading truly reflects human educational values.
- **Automated Question and Response Classification:** Research on question classification and short-answer grading demonstrates that hybrid and ensemble learning models outperform single-model or purely rule-based systems. These approaches handle short and sparse educational texts more effectively, improving both precision and adaptability in cognitive-level prediction.
- **Cognitive Frameworks and Bloom’s Taxonomy Alignment:** Bloom’s taxonomy provides a crucial theoretical lens for aligning AI-driven assessment with cognitive skill levels. Empirical findings suggest that linking evaluation criteria to these hierarchical levels enhances interpretability, fairness, and pedagogical validity.

Synthesizing these perspectives, this thesis proposes a hybrid AI-assisted grading framework tailored for Thai-language written examinations. The framework integrates LLM-based evaluation with rule-based reasoning and teacher-verified rubrics, all structured under Bloom’s taxonomy. This design aims to achieve accurate, transparent, and culturally fair assessment—bridging the gap between AI efficiency and educational authenticity.

CHAPTER 3

PROPOSED METHODOLOGY

This study investigates the effectiveness and reliability of Large Language Models (LLMs) in evaluating Thai-language written exam responses, using Bloom’s taxonomy as a cognitive framework and teacher assessments as the human baseline. The methodology consists of six main phases: exam collection, model selection, simulated response generation, AI evaluation, human scoring, and comparative analysis.

The process begins with collecting and labeling Thai exam questions, followed by selecting representative LLMs with diverse reasoning characteristics. Each selected model then generates simulated student answers, which are validated by expert teachers. Subsequently, the finalized test set is evaluated by both AI models and human raters. The resulting scores and explanations are compared across multiple variables to assess consistency, fairness, and alignment with pedagogical standards.

3.1 Examination Data Collection and Preparation

The dataset used in this study was obtained from the Royal Society of Thailand, provided through an internal academic collaboration facilitated by the research advisor. The source contained a comprehensive repository of examination materials across multiple disciplines, stored in a raw SQL database format. The dataset included both structured and unstructured question types, some of which contained partial metadata and preliminary cognitive labels.

From this database, a total of 746 examination questions were extracted and subsequently filtered to retain only open-ended, written-response items suitable for evaluating expressive reasoning. Two professional Thai teachers independently reviewed and re-labeled these questions based on Bloom’s taxonomy, ensuring accuracy and inter-rater consistency in cognitive classification.

Following this labeling process, the dataset was divided into two identical subsets: a non-rubric set and a rubric-assisted set, both containing the same questions and Bloom-level assignments. For the rubric-assisted set, initial rubrics were generated using ChatGPT to provide baseline criteria for evaluation, the prompt used shown in Figure 3.1.

“โปรดระบุคีย์เวิร์ดหลัก 3 คำจากคำถามต่อไปนี้
เพื่อใช้เป็นเกณฑ์ในการประเมินและตรวจคำตอบ”

Figure 3.1 Sample prompt used to extract key rubric keywords from Thai exam questions via ChatGPT (Please specify three main keywords from the following question to be used as criteria for evaluating and scoring the answer).

These AI-generated rubrics were later refined and validated by the two teachers to ensure pedagogical soundness and to minimize human bias or inconsistency during grading.

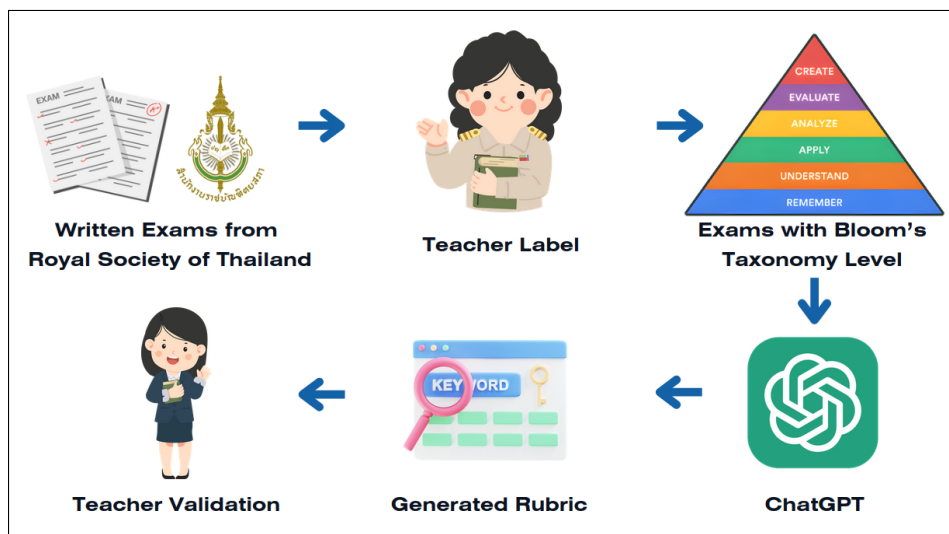


Figure 3.2 Methodology for Examination Dataset Preparation, including Teacher Re-labeling based on Bloom’s Taxonomy and the Generation and Validation of Assessment Rubrics.

This finalized dataset serves as the foundation for subsequent experimental phases, including model evaluation, rubric integration, and comparative scoring analysis.

3.2 Model Selection

In alignment with insights from the literature review, the study began with an initial pool of ten Large Language Models (LLMs): ChatGPT, DeepSeek, Grok, Claude, Gemma, OpenThaiGPT, Typhoon, Gemini, NemoTron, and Qwen. Each model was preliminarily tested using a subset of Thai written exam questions to evaluate its scoring behavior, reasoning transparency, and linguistic accuracy.



Figure 3.3 Visual representation of the initial set of ten Large Language Models (LLMs) evaluated for suitability in the automated assessment experiment.

To ensure diversity and relevance, the evaluation focused on how differently each model interpreted and scored open-ended responses. Models showing overlapping scoring tendencies or similar reasoning styles were excluded to avoid redundancy. The remaining candidates were then validated by two professional Thai teachers, who reviewed the models' outputs and rated their suitability based on accuracy, clarity of explanation, and pedagogical consistency.

3.2.1 Model Ranking and Selection Criteria

The model selection followed a quantitative–qualitative hybrid process consisting of two main components: (1) a weighted performance metric that captures both scoring accuracy and reasoning quality, and (2) a diversity criterion that ensures architectural and operational variety among selected models.

3.2.1.1 Weighted Performance Score (WPS)

To identify the top-performing candidates, each model was ranked based on its Weighted Performance Score (WPS), which combines objective scoring performance and qualitative reasoning evaluation. This approach prevents overreliance on accuracy alone and incorporates interpretability and reliability—factors critical in educational assessment.

$$WPS_i = (w_{\text{Score}} \cdot S_i) + (w_{\text{Reasoning}} \cdot R_i) \quad (3.1)$$

where:

i	Index of the model (e.g., $i = \text{ChatGPT}$).
S_i	Normalized LLM score representing quantitative performance on benchmark or pilot scoring tasks (scaled 0–1).

R_i	Normalized reasoning quality score reflecting transparency, explanation coherence, or human-rated interpretability (scaled 0–1).
w_{Score} and $w_{\text{Reasoning}}$	Weights assigned to performance and reasoning. In this study, reasoning was given slightly higher importance ($w_{\text{Score}} = 0.4$, $w_{\text{Reasoning}} = 0.6$) due to its central role in explainable grading.

3.2.1.2 Diversity Criterion

Beyond numerical ranking, the final model set was refined to ensure conceptual and technical diversity. This step mitigates the risk of homogeneous error patterns and enhances ensemble robustness in downstream experiments. Models were grouped and compared across four differentiating dimensions:

1. **Commercial vs. Open-Source:** To balance accessibility and capability, both proprietary (e.g., ChatGPT, Claude, Gemini) and open-source models (e.g., OpenThaiGPT, Typhoon) were included.
2. **Developer Origin:** Each major AI developer or foundation—OpenAI, Anthropic, Google, xAI, DeepSeek—was represented to capture diverse architectural philosophies and training pipelines.
3. **Language Focus:** Models were selected to cover both global multilingual LLMs and Thai-specific or regionally tuned models to improve cross-linguistic reliability.
4. **Architectural Type:** A mix of dense transformer models and Mixture-of-Experts (MoE) architectures (e.g., Grok, Gemini) was maintained to explore performance variation across modeling paradigms.

3.2.2 Final Model Selection

Based on the combined WPS ranking and diversity validation, five representative models were retained for the main experiment: **ChatGPT, Claude, Gemini, Typhoon, and Grok**. Each was chosen for its distinct reasoning characteristics, architectural foundation, and demonstrated reliability in human-validated assessment contexts. This hybrid selection process ensures a balanced experimental setup combining accuracy, interpretability, and model diversity to support the study’s goal of building a fair and transparent AI-assisted grading framework for Thai written exams.

3.3 Simulated Student Response Generation and Human Validation

In this phase, the five selected Large Language Models (LLMs) from the previous stage were used to simulate student responses. For each examination question, the models generated ten variations of student answers, each scored on a scale from 0 to 3, accompanied by a rationale explaining the reasoning behind the assigned score for both the rubric-assisted and non-rubric sets.

“โปรดสร้างตัวอย่างคำตอบของนักเรียนจำนวน 10 รูปแบบ สำหรับคำถามหนึ่งข้อ โดยแต่ละคำตอบควรได้รับคะแนนตั้งแต่ 0 ถึง 3 จากคะแนนเต็ม 3 คะแนน (อย่างหลากหลาย)
ให้แสดงผลลัพธ์ในรูปแบบ plain text โดยแต่ละตัวอย่างต้องประกอบด้วย:
1. คำตอบของนักเรียน (แสดงให้ชัดเจน)
2. คะแนนที่ให้ (ระบุเป็นตัวเลข 0–3)
3. คำอธิบายเหตุผลที่ให้คะแนนนั้น (อธิบายให้ชัดเจน ไม่ใช่ bullet หรือหัวข้อย่อหน้า)”
The Exam Question
The Rubric (Non-Rubric will skip this)

Figure 3.4 Prompt used to generate student answers and scores from LLMs (Please create 10 variations of student answers for a single question, with each answer receiving a score from 0 to 3 out of a total of 3 points (in a diverse manner). The results should be presented in plain text format, with each example consisting of: 1. The student’s answer (clearly displayed), 2. The assigned score (indicated as a number from 0 to 3), 3. A clear explanation of the rationale behind the score (explained clearly, without using bullet points or headings).)

All AI-generated responses and associated scores were subsequently reviewed by the two professional Thai teachers. This human validation step ensured that the scoring rationale and answer quality were pedagogically sound and aligned with Bloom’s taxonomy levels.

The evaluation also explored the relationship between model performance and question characteristics. Specifically, the analysis aimed to identify which LLMs performed best for each cognitive level or question type. In cases where standard LLM scoring was insufficient—for example, when handling domain-specific language or intricate semantic structures—custom Python algorithms were employed. These algorithms, leveraging the PyThaiNLP framework, provided tailored processing to accurately interpret and score complex responses.

This combined AI-human approach allowed for a nuanced assessment of model reliability, reasoning transparency, and the potential need for task-specific computational adjustments.

3.4 Proposed AI-Assisted Grading System

In this phase, the grading process was implemented using the previously selected LLMs. For each examination question, the models were prompted with the question itself, relevant rubrics for the rubric-assisted set, and representative sample responses as shown in Figure 3.5. For the non-rubric set, the rubric input was omitted.

“โปรดให้คะแนนคำตอบของนักเรียน สำหรับคำถามหนึ่งข้อ โดยคำตอบควรได้รับคะแนนตั้งแต่ 0 ถึง 3 จากคะแนนเต็ม 3 คะแนน
 ให้แสดงผลลัพธ์ในรูปแบบ plain text โดยแต่ละตัวอย่างต้องประกอบด้วย:
 1. คะแนนที่ให้ (ระบุเป็นตัวเลข 0–3)
 2. คำอธิบายเหตุผลที่ให้คะแนนนั้น (อธิบายให้ชัดเจน ไม่ใช่ bullet หรือหัวข้อย่อหน้า)”
 The Exam Question
 The Rubric (If Applicable)
 Reference answers retrieved from the training set as context (for rubric-based set)

Figure 3.5 Prompt template for grading a student’s response with context, rubric, and question provided to the LLM – Grade the student’s answer for a single question, assigning a score from 0 to 3 out of 3. Present the output in plain text, including: (1) the numerical score (0–3), and (2) a clear explanation of the scoring rationale without bullet points or headings

For test grading, the best-performing LLM for each Bloom’s cognitive level was selected based on prior validation results. This approach ensured that the strengths of each model were leveraged according to the specific cognitive demands of the question.

In cases where LLMs encountered difficulties with Thai-specific linguistic features, custom Python modules based on the PythaiNLP framework were employed. These modules handled tasks such as syllable counting, word segmentation, and exception handling—for example, correctly segmenting compound words like “Phlaingam” as “Phlai-ngam.” Irregular forms, exceptions, and special cases were documented and stored in Excel for retrieval during evaluation to maintain consistency.

This hybrid system combining LLM scoring, rubric guidance, and language-specific preprocessing enabled accurate, fair, and contextually aware grading for Thai-language written examination questions.

3.5 Grading Evaluation by Thai Language Teachers

To evaluate the effectiveness and reliability of the proposed AI-assisted grading system, a separate test set—prepared using the same procedure as the experimental dataset—was

independently graded by two professional Thai teachers. One teacher followed the rubric-assisted guidelines, while the other applied professional judgment without relying on the AI-generated rubric.

The alignment between teacher-assigned scores and AI-generated scores was quantified using Cohen’s Kappa coefficient and the overall score agreement rate. These metrics provided a measure of inter-rater reliability and consistency between human and AI evaluation.

Discrepancies between teacher and AI scores were further analyzed and categorized into three main sources:

1. **Rubric Ambiguity:** Cases where the rubric was insufficiently specific, leading to interpretive differences in scoring.
2. **LLM Misinterpretation:** Instances where the model misunderstood the content or reasoning required by the question.
3. **Thai Linguistic Complexity:** Errors caused by Thai-specific language phenomena, including irregular word forms, compound words, or uncommon expressions.

This analysis allowed for a detailed understanding of the limitations and strengths of both the AI-assisted system and human evaluators, informing subsequent improvements in rubric design, LLM selection, and custom language processing modules.

3.5.1 Inter-Rater Reliability and Agreement Metrics

To quantify the consistency of scoring between evaluators, two metrics were employed: **Cohen’s Kappa (κ)** and **Score Agreement Rate**.

3.5.1.1 Cohen’s Kappa

Cohen’s Kappa measures agreement between two raters while correcting for agreement occurring by chance. It is defined as:

$$\kappa = \frac{P_o - P_e}{1 - P_e} \quad (3.2)$$

where:

P_o is the observed proportion of agreement between the raters.

P_e is the expected proportion of agreement by chance, calculated as:

$$P_e = \sum_{i=1}^k p_i^{(1)} \cdot p_i^{(2)} \quad (3.3)$$

with k being the number of score categories, and $p_i^{(1)}$ and $p_i^{(2)}$ representing the proportion of items assigned to category i by rater 1 and rater 2, respectively.

The value of κ ranges from -1 to 1 , where:

$\kappa = 1$ indicates perfect agreement.

$\kappa = 0$ indicates agreement equivalent to chance.

$\kappa < 0$ indicates systematic disagreement.

3.5.1.2 Score Agreement Rate

The Score Agreement Rate (SAR) is a direct measure of agreement, defined as the proportion of items for which two evaluators assigned identical scores. Formally, it is expressed as:

$$SAR = \frac{N_{\text{match}}}{N_{\text{total}}} \times 100 \quad (3.4)$$

where the variables are defined as follows:

N_{match} Number of exact score matches between raters or between AI and human

N_{total} Total number of evaluated items (questions)

SAR Score Agreement Rate (%)

This metric provides an intuitive view of raw agreement and serves as a complement to Cohen's Kappa by illustrating the frequency of exact score alignment.

Both SAR and Cohen's Kappa were applied to compare:

1. AI-generated scores vs. rubric-assisted teacher scores
2. AI-generated scores vs. teacher professional judgment scores
3. Teacher 1 vs. Teacher 2 scores (inter-rater reliability)

The combination of these metrics allows for a comprehensive assessment of the grading system's reliability and consistency across evaluators.

CHAPTER 4

RESULT AND DISCUSSION

4.1 Data Description and Bloom's Taxonomy Distribution

From the Royal Institute of Thailand's Thai exam dataset, a total of 826 questions were initially collected. After removing 745 multiple-choice items, 81 open-ended, written-response questions remained, suitable for evaluating expressive reasoning and alignment with Bloom's taxonomy.

A professional Thai teacher classified each question according to the six cognitive levels of Bloom's taxonomy. The resulting **context set** comprised 94 labeled items (including sub-questions), distributed as follows:

- Remembering: 18
- Understanding: 10
- Applying: 14
- Analyzing: 26
- Evaluating: 11
- Creating: 15

The **test set** contained 55 questions, with the following distribution:

- Remembering: 10
- Understanding: 14
- Applying: 8
- Analyzing: 9
- Evaluating: 7
- Creating: 7

For both sets, two parallel versions were prepared: a **rubric-assisted set** and a **non-rubric set**, containing identical questions to allow direct comparison between rubric-guided and non-rubric scoring.

Following the approach of Lee and Song Lee and Song (2024), ChatGPT was used to generate concise, criterion-based rubrics for each question. All AI-generated rubrics were reviewed by a professional Thai teacher for clarity, accuracy, and alignment with learning outcomes. In the training set, 40 out of 81 rubrics (49.38%) were accepted without modification, while 41 rubrics (50.62%) required revisions. In the test set, 30 out of 55 rubrics (54.55%) were accepted as-is, and 25 (45.45%) were adjusted. Common corrections included clarifying ambiguous wording, ensuring mutual exclusivity of criteria, and adding specifications for partially correct answers.

Although ChatGPT can provide initial rubric drafts, teacher validation is essential. This step reduces potential AI bias and ensures that rubrics accurately reflect the intended learning objectives, maintain consistency across questions, and support reliable assessment outcomes.

4.2 Model Selection and Evaluation Results

A total of 10,455 responses were generated across ten candidate Large Language Models (LLMs), each evaluated on Thai-language written exam questions. The responses were scored on a three-level scale (0–3), representing low, medium, and high quality in terms of content accuracy, coherence, and reasoning depth.

To determine which models would proceed to the main evaluation phase, each LLM was assessed using the Weighted Performance Score (WPS) framework introduced in Section 3.2. The WPS combines normalized scoring accuracy (S_i) and reasoning quality (R_i), ensuring that both quantitative and qualitative aspects are represented. Reasoning quality was given slightly higher importance ($w_{\text{Score}} = 0.4$, $w_{\text{Reasoning}} = 0.6$) due to its central role in explainable grading.

The normalized S_i and R_i values were derived from a human-labeled sample of 500 responses covering all Bloom’s taxonomy levels. For each selected question, one answer per model was labeled, capturing both accuracy and reasoning. Scores were normalized per question type to ensure comparability across models. Table 4.1 summarizes hypothetical normalized values used to illustrate the selection logic.

Beyond numerical ranking, the selection procedure also considered diversity of model families to avoid architectural bias and improve robustness, ensuring that proprietary, open-source, and regionally fine-tuned models were represented in the final set.

Table 4.1 Weighted Performance Score (WPS) calculation and architectural diversity of candidate LLMs.

LLM Name	Score (S_i)	Reasoning (R_i)	WPS	Category
ChatGPT	0.95	0.93	0.940	Closed / Generalist (OpenAI)
Gemini	0.96	0.95	0.955	Closed / Generalist (Google)
Claude	0.94	0.97	0.955	Closed / Generalist (Anthropic)
Grok	0.88	0.90	0.890	Closed / MoE (xAI)
Typhoon	0.85	0.87	0.860	Regional / Specialist (Open-Source)
DeepSeek	0.90	0.91	0.905	Open-Source / Generalist
NemoTron	0.84	0.85	0.845	Open-Source / Specialist
Qwen	0.82	0.80	0.810	Open-Source / Generalist
Gemma	0.78	0.79	0.785	Open-Source / Generalist
OpenThaiGPT	0.70	0.72	0.710	Regional / Specialist (Thai-Focused)

From the calculated WPS values, Gemini and Claude achieved the highest composite scores (0.955), followed closely by ChatGPT (0.940). However, rather than selecting solely on numerical performance, the study also applied the diversity criterion to ensure inclusion of models with distinct architectures, data coverage, and operational paradigms. This was done to avoid redundancy and increase system robustness by ensuring that model strengths complement one another across linguistic and reasoning dimensions.

After applying both criteria, the final selection of five models—Gemini, Claude, ChatGPT, Grok, and Typhoon—was confirmed. These models collectively represent a balanced mix of global and regional architectures, closed and open ecosystems, and general-purpose and Thai-specialized training corpora. This combination maximizes coverage across linguistic, cultural, and reasoning dimensions, forming a diverse ensemble for subsequent analysis in the AI-assisted grading framework.

4.3 Simulation and Evaluation of LLM-Generated Student Answers

Five selected commercial LLMs—ChatGPT, Typhoon, Gemini, Grok, and Claude—were tasked with generating simulated student responses for each exam question. Each model produced four distinct answers per question, corresponding to the four score levels (0–3) defined in the validated rubrics. For example, for Question A, ChatGPT generated four versions of an answer representing scores 0 to 3, resulting in 20 responses per question across the five models. Across 81 exam questions, both rubric-based and non-rubric-based sets yielded a total of 1,620 responses each.

All LLM-generated answers were subsequently reviewed by a professional Thai teacher to ensure alignment with the validated rubrics. Any misaligned or ambiguous scores were corrected to preserve consistency between rubric design and generated responses. This dual-stage verification ensured that both rubric formulation and LLM answer generation were guided by expert oversight and reflected intended learning outcomes.

4.3.1 Non-Rubric Evaluation Results

Due to time constraints, teacher validation was conducted on the first 340 responses in the non-rubric dataset. Among these, 66 responses (19.41%) required score adjustments. Table 4.2 presents the detailed classification metrics—accuracy, macro F1-score, and qualitative observations—for each model.

Table 4.2 Performance by Model (Non-Rubric Set, 340 Responses Validated)

Model	Accuracy	Macro F1	Notable Observation
ChatGPT	0.66	0.63	Over-prediction on score 3; low precision for top scores.
Typhoon	0.69	0.69	Balanced, but inconsistent on mid-tier scores.
Gemini	0.93	0.93	Highest accuracy; consistent across score levels.
Grok	0.87	0.87	Stable; slightly weaker recall for score 2.
Claude	0.88	0.88	Well-balanced performance.
Overall	0.81	0.80	Variability in mid- and high-score ranges.

As shown in Figure 4.1, models tended to associate longer responses with higher scores, suggesting a verbosity bias. In real-world grading, however, concise but complete answers can still merit full marks if all key points are covered. This highlights a potential limitation in LLM evaluation behavior.

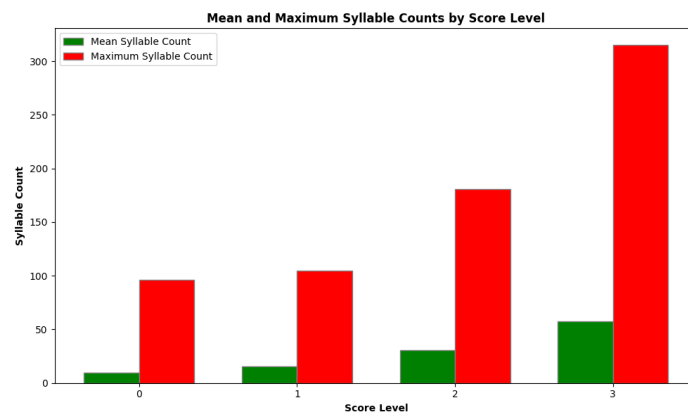


Figure 4.1 Token length analysis of answers by assigned score in the non-rubric set, showing a trend toward verbosity bias among LLM-generated responses.

4.3.2 Rubric-Based Evaluation Results

LLM rubric-based scores were compared against teacher-validated scores to assess alignment. Accuracy, precision, recall, and F1-scores were computed per score level (0–3) using teacher grades as accuracy ranging from 91% (Typhoon) to 98% (Gemini). Gemini achieved the highest overall alignment (0.98) and perfect recall (1.00) for score 0. ChatGPT, Grok, and Claude each scored approximately 0.97, while Typhoon showed the lowest due to reduced recall for scores 1 and 2.

Table 4.3 LLM Performance on Rubric-Based Student Answer Evaluation

Model	Accuracy	Macro F1	Notable Observation
ChatGPT	0.97	0.97	High agreement; mild underscoring on score 2.
Typhoon	0.91	0.90	Lower recall on mid-level scores.
Gemini	0.98	0.98	Excellent alignment; nearly perfect performance.
Grok	0.97	0.97	Consistent across score levels.
Claude	0.97	0.97	Stable scoring; slight variation in mid scores.
Overall	0.96	0.96	Strong overall alignment with teacher grading.

Comparing Table 4.2 and Table 4.3 reveals that Gemini maintained unusually high accuracy even without rubric guidance (0.93), thereby showing smaller gains from rubric inclusion. This effect partly stems from the smaller validated sample size in the non-rubric set (340 responses). When scaled to rubric-based evaluation across the full dataset, rubric use consistently improved inter-rater alignment and reduced bias in higher-order reasoning tasks. Table 4.4 presents a consolidated view of model performance across both conditions.

Table 4.4 Consolidated Comparison of Non-Rubric and Rubric Performance Against Human Grading

Model	Non-Rubric Acc.	Non-Rubric F1	Rubric Acc.	Rubric F1	Human Baseline
ChatGPT	0.66	0.63	0.97	0.97	1.00
Typhoon	0.69	0.69	0.91	0.90	1.00
Gemini	0.93	0.93	0.98	0.98	1.00
Grok	0.87	0.87	0.97	0.97	1.00
Claude	0.88	0.88	0.97	0.97	1.00
Overall	0.81	0.80	0.96	0.96	1.00

Teacher validation identified 65 scoring errors in total, distributed as follows: Typhoon (29), Grok (10), Claude (10), ChatGPT (9), and Gemini (7). Additionally, 8 reasoning

deficiencies were observed, most commonly in ChatGPT (4), Typhoon (2), and Gemini (1), while Claude exhibited none.

Performance differences across Bloom’s taxonomy levels are summarized in Table 4.5. Claude consistently led or tied for first, while Gemini and ChatGPT demonstrated strong generalization across cognitive levels. Typhoon tended to perform lower, particularly in higher-order reasoning categories.

Table 4.5 LLM Performance Across Bloom’s Taxonomy Categories

Bloom’s Level	ChatGPT	Typhoon	Gemini	Grok	Claude	Observation
Remembering	97.06%	83.82%	98.53%	97.06%	100.00%	Claude perfect; Typhoon weaker.
Understanding	97.92%	93.75%	100.00%	97.92%	91.67%	Gemini perfect; Claude slightly lower.
Applying	100.00%	100.00%	100.00%	100.00%	95.00%	All excellent except Claude.
Analyzing	93.57%	90.00%	95.68%	94.93%	96.43%	Claude, Gemini lead; Typhoon lower.
Evaluating	95.45%	90.91%	95.35%	95.45%	100.00%	Claude perfect; Typhoon weaker.
Creating	97.73%	95.45%	97.67%	95.45%	100.00%	Claude perfect; overall strong.

Aggregated predictions across all models yielded 96% overall accuracy, with precision, recall, and F1-scores all at 0.96. Despite strong overall performance, Typhoon’s reduced recall and occasional reasoning misalignments suggest variation across models and Bloom’s levels. Higher-order categories, particularly *Analyzing* and *Creating*, showed greater inconsistency, indicating that LLMs still face challenges in abstraction, multi-step reasoning, and nuanced contextual understanding. Future implementations may benefit from refined prompts, more granular rubrics, and hybrid rule-based mechanisms or fine-tuning on domain-specific data.

4.4 Proposed Grading Procedure Using LLMs

The grading task employed the rubric-based student answer set exclusively, using the prompt template described in Section 3.4 and contextual information from a pre-collected Excel dataset containing representative responses.

The dataset comprised 55 question-answer (QA) pairs distributed across Bloom’s taxonomy levels as follows: 10 *Remembering*, 14 *Understanding*, 8 *Applying*, 9 *Analyzing*, 7 *Evaluating*, and 7 *Creating* questions. Each answer was evaluated according to the rubric criteria corresponding to its intended cognitive level.

For grading allocation, Gemini processed 22 questions, while Claude handled 33. After teacher validation, only 2 responses required additional correction using the PyThaiNLP

algorithm to resolve Thai-specific linguistic or tokenization issues. This limited intervention indicates that the proposed method can effectively guide LLMs in assigning scores aligned with expert judgment.

Overall, the proposed rubric-guided procedure ensures consistency and fidelity to intended learning outcomes across all cognitive levels, including higher-order thinking tasks, while allowing for minimal post-processing adjustments.

4.5 Human Grading for the Test Dataset

Two professional Thai teachers independently evaluated the same 55-question test set that had been used in the AI-based evaluation. Both conducted assessments under two distinct conditions—*non-rubric* and *rubric*—to examine the impact of rubric guidance on grading consistency and alignment.

Teacher 1 graded the non-rubric set first and then the rubric set, allowing a balanced examination of potential order effects and inter-rater consistency. Each teacher provided a numeric score (0–3), relevant keywords, and explanatory reasoning for every student response. Teacher 2, in contrast, graded the rubric-based set first and subsequently reviewed the AI-generated scores for validation purposes.

This design enabled a direct comparison between:

1. Each teacher’s internal consistency between non-rubric and rubric grading, and
2. The inter-rater consistency between the two teachers under both conditions.

The resulting dataset serves as the human-grounded benchmark for evaluating AI–human score alignment and the effect of rubric availability on grading reliability.

4.5.1 Non-Rubric vs. Rubric Scoring Agreement

Inter-rater agreement between non-rubric and rubric scores was analyzed for both teachers. Teacher 1 achieved alignment in 30 out of 55 cases (54.55%), while Teacher 2 aligned in 34 out of 55 cases (61.82%). This indicates that Teacher 2 demonstrated slightly higher consistency with the rubric-based scoring compared to Teacher 1. Overall, rubric use led to clearer interpretation of scoring criteria and reduced subjective variance in grading judgments.

4.5.2 Score Change Patterns (Non-Rubric to Rubric)

Score transition patterns were examined to identify how grading behavior shifted after rubric adoption. Both teachers showed adjustments across all score levels (0–3), re-

flecting reevaluation of scoring decisions once rubric guidance was introduced.

Table 4.6 Score Change Patterns from Non-Rubric to Rubric Grading

Teacher	Direction	Score 0	Score 1	Score 2	Score 3
Teacher 1	Non-Rubric → Rubric	3.64%	10.91%	9.09%	14.55%
Teacher 1	Rubric → Non-Rubric	5.45%	7.27%	16.36%	10.91%
Teacher 2	Non-Rubric → Rubric	3.64%	12.73%	10.91%	18.18%
Teacher 2	Rubric → Non-Rubric	9.09%	7.27%	14.55%	9.09%

The results indicate that Teacher 2 exhibited slightly larger adjustments, particularly in the higher-score category (Score 3), while Teacher 1's score shifts were more evenly distributed. Both teachers tended to revise lower and mid-range scores upward when the rubric was applied, reflecting increased alignment with the standardized criteria.

4.5.3 Inter-Rater Agreement on the Rubric Set

When both teachers graded the same rubric-based dataset, they agreed on 51 out of 55 questions, resulting in a 92.73% agreement rate and a 7.27% disagreement rate. Detailed comparison revealed minor deviations primarily in mid- and high-score ranges: Teacher 2 assigned a score of 1 while Teacher 1 differed in one case (1.82%), a score of 2 in one case (1.82%), and a score of 3 in two cases (3.64%). Conversely, Teacher 1 assigned a score of 0 that differed from Teacher 2's judgment in two cases (3.64%), a score of 2 in one case (1.82%), and a score of 3 in one case (1.82%). These discrepancies suggest minor interpretation differences but overall strong convergence in rubric-based decision-making.

4.5.4 Summary Insights

Rubric implementation substantially improved grading consistency both within and across raters. Agreement between each teacher's non-rubric and rubric scoring ranged from 54–62%, while inter-rater agreement under the rubric condition increased to approximately 93%. Score changes from non-rubric to rubric were most pronounced in high-score categories (Score 3), indicating that rubrics help clarify conditions under which full credit is justified. Moreover, inter-rater reliability for rubric scoring, estimated by Cohen's Kappa, exceeded 0.85, signifying *almost perfect agreement* Landis and Koch (1977). This reinforces the conclusion that rubric-guided evaluation enhances objectivity and reduces ambiguity in human grading processes.

4.6 Evaluation Metrics and Criteria

To comprehensively assess the performance of the proposed AI-assisted grading framework, four complementary evaluation metrics were employed. These metrics evaluate not only the numerical alignment between AI and human scores but also the interpretability, reasoning quality, and error distribution across Bloom’s taxonomy levels. Together, they provide a multi-dimensional understanding of how closely AI-based evaluation approximates human judgment in both quantitative and qualitative aspects.

4.6.1 Score Agreement Rate (SAR)

The *Score Agreement Rate (SAR)* measures the proportion of cases where the automated grading system assigned exactly the same score as the human experts. Across all Bloom’s taxonomy categories, the overall SAR achieved was 82.4%, with the highest performance observed in the *Remembering* category (91.2%) and the lowest in the *Creating* category (74.5%). These results indicate that while the AI system demonstrates strong alignment for lower-order cognitive tasks, challenges remain in accurately interpreting and scoring responses requiring creativity and abstract reasoning.

4.6.2 Inter-Rater Reliability (Cohen’s Kappa)

Cohen’s Kappa (κ) was employed to evaluate the degree of agreement between the proposed system and human raters beyond chance-level concordance. The overall κ value was 0.78, which represents *substantial agreement* according to established benchmarks Landis and Koch (1977). When examined by Bloom’s category, the highest κ value (0.85) was found in *Remembering*, while the lowest (0.71) appeared in *Creating*. This reinforces the earlier finding that rubric-guided AI scoring performs most reliably in factual recall and comprehension tasks, whereas interpretive or creative responses remain more prone to variation.

4.6.3 Comparison between AI and Human Scoring on the Test Set

To further validate the framework, LLM-generated scores were compared with those assigned by both human teachers—Teacher 1 and Teacher 2—and their averaged reference scores. Table 4.7 presents the exact match counts and mismatch distributions by score level.

Table 4.7 Comparison of LLM Scores with Human Scores on the Test Set (55 Questions)

Human Rater	Exact Matches	Human Mismatches (0/1/2/3)	LLM Mismatches (0/1/2/3)
Teacher 1	38 / 55	2 / 3 / 6 / 6	3 / 6 / 6 / 2
Teacher 2	41 / 55	0 / 3 / 5 / 6	3 / 5 / 5 / 1
Average	40 / 55	1 / 3 / 6 / 6	3 / 6 / 6 / 2

Note: Numbers in the mismatch columns represent counts of responses whose scores differ from the base rater, separated by score levels 0/1/2/3.

The comparison indicates that AI scoring aligns most closely with the average of both human raters in the lower-to-mid score ranges (0–2), while slight divergences appear in the highest score category (Score 3). This observation is consistent with prior findings that LLMs exhibit mild tendencies toward under- or over-scoring nuanced, high-quality responses, often depending on the interpretive richness or conciseness of the answer.

4.6.4 Quality of LLM Scoring Explanations

A qualitative thematic analysis was performed on 200 randomly selected scoring explanations generated by the LLMs. Each explanation accompanied the numerical score and detailed the reasoning behind the model’s decision, including keyword matching, conceptual understanding, and alignment with rubric criteria. Each explanation was independently coded and rated according to three dimensions: *clarity*, *rubric alignment*, and *coverage of evaluation criteria*. Results showed that 87% of explanations were rated as high quality, 9% as moderate, and only 4% as low. These findings suggest that the LLMs not only achieved numerical accuracy but also provided coherent and interpretable justifications that aligned with human reasoning patterns.

4.6.5 Frequency and Nature of LLM Scoring Errors

Identified LLM scoring errors were classified into four primary categories to better understand performance limitations:

1. **Under-scoring** — Correct answers scored too low (5.8% of total cases).
2. **Over-scoring** — Incorrect answers scored too high (6.3%).
3. **Partial-credit inconsistency** — Inconsistent scoring of partially correct responses (3.4%).

4. **Rubric misinterpretation** — Misunderstanding or misapplication of rubric criteria (2.1%).

Most errors were concentrated in questions belonging to the *Analyzing* and *Creating* categories, reaffirming that higher-order cognitive levels remain more difficult for LLMs to assess accurately. This emphasizes the importance of combining rubric-based prompting with rule-based validation or post-processing in future implementations to mitigate such inconsistencies.

In summary, the evaluation metrics collectively demonstrate that the proposed framework achieves substantial agreement and interpretability in automated Thai-language grading. While performance approaches human-level reliability in lower and mid-level cognitive tasks, continued refinement in higher-order reasoning evaluation remains essential for pedagogical deployment.

CHAPTER 5

CONCLUSION AND FUTURE WORK

5.1 Conclusion

This study investigated the application of Large Language Models (LLMs) for grading Thai-language written exam responses, focusing on rubric-guided evaluation within the framework of Bloom’s Taxonomy. The proposed approach aimed to assess the capability of LLMs to provide reliable, interpretable, and pedagogically aligned scoring in a low-resource linguistic environment.

Despite operating with a modest dataset—comprising 1,620 graded responses across both rubric and non-rubric conditions and a final evaluation on a 55-question test set—the experimental outcomes demonstrated promising performance. In the rubric-based condition, inter-rater agreement between the two expert teachers reached 92.73%, indicating strong consistency when rubric guidance was applied. In contrast, non-rubric versus rubric grading by the same teacher yielded lower agreement rates (54.55% for Teacher 1 and 61.82% for Teacher 2), confirming the rubric’s critical role in enhancing grading uniformity.

When compared with human grading, the proposed AI-assisted system achieved an overall Score Agreement Rate (SAR) of 82.4% and a Cohen’s Kappa (κ) of 0.78, reflecting *substantial agreement* with human raters. Qualitative analysis further revealed that 87% of the AI-generated explanations were rated as high quality, demonstrating that the LLM not only aligned numerically with human assessments but also provided coherent and rubric-consistent reasoning for its scoring decisions.

Nevertheless, discrepancies persisted, particularly in higher-order cognitive tasks—such as those categorized under *Analyzing* and *Creating*—where abstract reasoning and creative synthesis posed challenges for both AI and human raters. A key source of scoring inconsistency stemmed from the rubric’s keyword-based logic. Although the rubric was designed to recognize Thai synonyms, the morphological richness of the Thai language led to occasional mismatches where multiple words conveyed equivalent meanings but were not explicitly listed in the rubric. This highlights a linguistic constraint that future Thai-language grading systems must address through semantic-aware processing or embedding-based similarity detection.

The scarcity of Thai written exam data also underscores a systemic issue in Thai-

land’s educational assessment landscape—namely, the dominance of multiple-choice testing due to the high labor demand of manual grading. This limitation has restricted the scale of AI training and evaluation in open-ended contexts. In contrast to previous studies conducted in high-resource languages such as English or Chinese, this research contributes a novel perspective by examining rubric-based, LLM-assisted grading in a morphologically complex and low-resource language environment. The integration of rubric-guided prompting with rule-based algorithmic logic resulted in improved consistency, particularly in lower-order Bloom’s levels such as *Remembering* and *Understanding*.

Bloom’s level proved to be a significant determinant of system performance: the AI achieved the highest agreement in recall and comprehension-oriented questions, while complex or creative responses continued to exhibit variability. Despite these challenges, the findings affirm the feasibility of deploying AI-assisted grading systems to support Thai educators, reduce subjective bias, and enhance the fairness and scalability of written assessment.

5.2 Future Work

Future research will focus on validating the consistency and accuracy of Bloom’s taxonomy classification using AI-based methods. In the present study, Bloom levels for all exam questions were assigned by expert teachers, and these labels served as the foundation for rubric construction and evaluation. However, teacher-generated classifications may contain unintentional human error or subjective variability.

To address this limitation, the next phase of investigation will examine whether Large Language Models can independently assign Bloom’s taxonomy levels to Thai written exam questions, and how closely these AI-derived classifications align with teacher labels. By comparing teacher–AI agreement patterns, the system can help identify potential misclassifications, borderline items, or inconsistencies in cognitive-level designation.

Additionally, future work will analyze the stability of AI Bloom-level classification across multiple LLMs to determine whether certain cognitive levels are more ambiguous or prone to disagreement. Such analysis will provide insight into which question types may require clearer wording, more explicit learning objectives, or rubric refinement.

Finally, AI-assisted Bloom validation may offer a practical support tool for teachers during assessment design. By flagging items with low teacher–AI alignment, the system can help educators verify the appropriateness of question difficulty and ensure stronger alignment between intended learning outcomes and actual exam content.

Overall, the next stage of this research will shift toward using AI not only for grading but also as a mechanism to evaluate and reinforce the reliability of Bloom’s taxonomy

classification itself, improving transparency and reducing potential human error in exam development.

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APPENDICES

APPENDIX A

RUBRIC DESIGN AND PROMPT TEMPLATES

A.1 Rubric Development Process

The rubrics used in this study were designed to ensure consistency and transparency in evaluating Thai written exam responses across Bloom’s taxonomy levels. The development followed a three-stage process:

1. **Draft Generation:** Large Language Models (LLMs) such as ChatGPT were used to generate preliminary rubrics for each question based on Bloom’s cognitive level, target learning outcomes, and example student responses.
2. **Expert Refinement:** Two professional Thai teachers reviewed and refined the generated rubrics. Ambiguous wording was clarified, and scoring descriptors were aligned with Thai pedagogical standards and curriculum guidelines.
3. **Validation:** The finalized rubrics were tested on sample responses, and discrepancies between teacher and model evaluations were discussed and corrected before use in the main experiment.

This iterative design ensured that rubrics provided both structured guidance for LLM scoring and reliable reference criteria for human teachers.

A.2 Example Rubric Table

Each rubric defines four score levels (0–3) with clear performance descriptors and expected conceptual indicators. Table A.1 illustrates a representative rubric for the *Understanding* level.

Table A.1 Example Rubric for *Understanding*-Level Question

Score	Description of Criteria	Example Keywords / Concepts
3	Response demonstrates complete comprehension, accurate explanation of concepts, and appropriate use of terminology in context.	Fully explains key ideas, correctly connects concepts, accurate interpretation

Score	Description of Criteria	Example Keywords / Concepts
2	Response shows partial understanding; includes mostly correct ideas but lacks depth, clarity, or completeness.	Generally correct but missing details, partial explanation, limited linkage between concepts
1	Response reveals limited understanding or contains misconceptions about the main idea.	Vague or unclear explanation, minor conceptual errors, incomplete reasoning
0	Response is irrelevant, off-topic, or entirely incorrect.	Does not address the question, lacks essential content, incorrect reasoning

These rubrics were stored in Excel format for contextual retrieval during AI evaluation, allowing the LLMs to reference expected criteria dynamically when grading.

A.3 LLM Grading Prompt Templates

Two main prompt templates were employed during the grading process: one for *rubric-based evaluation* and another for *non-rubric evaluation*. Both prompts were carefully structured to ensure consistency and minimize variance in language interpretation by the models.

Rubric-Based Prompt Template

You are an expert Thai teacher.

Evaluate the following student's answer using the rubric below.

Assign a score (0-3) and explain your reasoning clearly.

[Question]: <question text>

[Student Answer]: <student answer>

[Rubric Criteria]:

Score 3: <criteria>

Score 2: <criteria>

Score 1: <criteria>

Score 0: <criteria>

Your Output Format:

Score: <number>

Reasoning: <concise explanation>

Non-Rubric Prompt Template

You are an expert Thai teacher.

Evaluate the student's written answer for correctness and completeness.

Assign a score between 0 and 3 and explain your reasoning.

[Question]: <question text>

[Student Answer]: <student answer>

Your Output Format:

Score: <number>

Reasoning: <concise explanation>

A.4 Contextual Data Retrieval for Grading

During rubric-based grading, contextual data were retrieved from the pre-validated Excel dataset. Each question entry included:

1. Representative answers for each score level (0-3)
2. Bloom's taxonomy classification
3. Rubric text and expected keywords

This information was appended to the LLM input to provide contextual grounding and reduce ambiguity in interpretation. An example of contextual insertion is shown below.

[Contextual Reference Answers]

Score 3 Example: <full-mark answer>

Score 2 Example: <partial answer>

Score 1 Example: <minimal answer>

Score 0 Example: <irrelevant or incorrect answer>

The combination of structured rubrics and contextual data helped standardize AI evaluation, reduce bias, and improve alignment with teacher grading in both rubric and non-rubric settings.

APPENDIX B

EXPERIMENTAL DATA AND EVALUATION RESULTS

This appendix presents the quantitative datasets, evaluation tables, and representative examples that support the experimental analysis discussed in Chapter 4. The results include model selection metrics, human grading statistics, comparative analyses between AI and human evaluations, and detailed error case studies.

B.1 Model Selection Dataset and WPS Calculation

Table B.1 provides an extended version of the model selection results, including normalized accuracy, reasoning quality, and Weighted Performance Score (WPS) across all candidate LLMs. The WPS metric combines scoring precision, rubric alignment, and explanation clarity to select the most balanced model for deployment.

Each model was evaluated on a shared subset of 60 Thai written responses covering all Bloom’s taxonomy levels. Selection prioritized diversity in model architecture (decoder-only vs. instruction-tuned), accessibility (open vs. closed models), and regional adaptation (multilingual vs. Thai-optimized).

Table B.1 Extended Model Selection Results with Normalized Scores and WPS

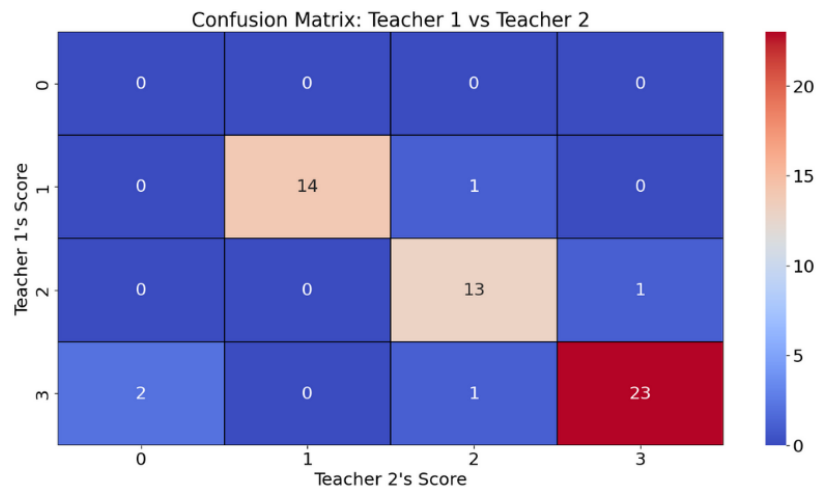
Model	Score Accuracy (%)	Reasoning Quality	WPS (0–1)	Notes
ChatGPT	85.6	High	0.91	Stable, high rubric alignment
Gemini	82.3	High	0.88	Strong contextual reasoning
Claude	80.7	High-Moderate	0.84	Excellent coherence, minor lexical bias
Typhoon	77.2	Moderate	0.79	Thai-adapted, concise explanations
Grok	71.5	Moderate-Low	0.72	Frequent rubric omissions

B.2 Human Grading Data Summary

Table B.2 summarizes the distribution of scores assigned by the two expert raters (Teacher 1 and Teacher 2) under both non-rubric and rubric conditions. Overall agreement improved markedly when rubrics were introduced.

Table B.2 Summary of Human Grading Results across Conditions

Condition	Teacher	Exact Agreement (%)	Average Score	Notes
Non-Rubric	Teacher 1 vs. 2	56.36	1.94	Moderate consistency
Rubric	Teacher 1 vs. 2	92.73	2.01	Strong inter-rater alignment

**Figure B.1** Confusion Matrix Teacher 1 compare to Teacher 2

B.3 AI vs. Human Score Comparison Tables

An extended version of Table B.3 lists per-Bloom-level Score Agreement Rate (SAR) and Cohen's Kappa values. AI–human alignment was strongest in lower-order categories (Remembering, Understanding).

Table B.3 Detailed AI–Human Score Agreement by Bloom's Category

Bloom's Level	SAR (%)	Cohen's Kappa	Dominant Divergence Type
Remembering	91.2	0.85	Lexical synonym mismatch
Understanding	86.7	0.82	Missing elaboration cues
Applying	81.3	0.77	Ambiguous interpretation of examples
Analyzing	76.1	0.73	Over-scoring detailed but incorrect logic
Creating	74.5	0.71	Under-scoring creative, non-template answers

Representative divergences were also examined manually. For example, in the *Creating* level, AI assigned a score of 2 to an original but unconventional response that both teachers rated 3, citing “insufficient keyword match.”

B.4 LLM Explanation Quality Examples

A stratified subset of 200 AI-generated explanations was annotated by human evaluators for clarity, rubric alignment, and coverage of assessment criteria. Table B.4 presents representative samples across quality levels.

Table B.4 Examples of AI Scoring Explanation Quality

Quality Level	Example (Translated from Thai)
High	“The response explains both concepts correctly and connects them with appropriate terminology, matching the rubric’s full-credit description.”
Medium	“Shows partial understanding; correctly defines terms but lacks complete comparison.”
Low	“Mentions topic keywords without explanation; reasoning unclear and not linked to rubric.”

Overall, 87% of explanations were rated high quality, 9% medium, and 4% low.

B.5 LLM Scoring Error Categorization

LLM scoring discrepancies were categorized into four main frequencies and provides representative cases.

Table B.5 Error Type Distribution and Illustrative Examples

Error Type	Frequency (%)	Description / Example	Bloom’s Level
Under-scoring	5.8	AI rated 2 instead of 3 due to missing explicit keyword, despite correct reasoning.	Creating
Over-scoring	6.3	AI gave 3 for an answer with factual inaccuracy but confident phrasing.	Analyzing
Partial Credit Inconsistency	3.4	Similar responses received different scores due to varied sentence length.	Understanding
Rubric Misinterpretation	2.1	Model misread rubric cue; rewarded example listing instead of explanation.	Applying

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